

Anti-Lock Brake System (ABS) and Stability Control



Special Tool(s)

 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST3030-A	Rotunda Active Wheel Speed Sensor Tester 420-119-KIT or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operations

Anti-Lock Braking

The ABS module continuously monitors brake pedal input, lateral vehicle motion and the rotational speed of each wheel. The PCM sends the brake pedal switch information to the ABS module over the HS-CAN while the RCM sends lateral acceleration sensor information to the ABS module over a private HS-CAN . Wheel speed information is retrieved by the ABS module using 4 active wheel speed sensors. When the ABS module detects an impending wheel lock during a braking event, the ABS module modulates brake pressure to the appropriate brake caliper(s) by opening and closing the appropriate solenoid valves inside the HCU while the hydraulic pump motor is activated. Once the affected wheel(s) return to the desired speed, the ABS module returns the solenoid valves in the HCU to their normal position.

The ABS module has 2 self-test options, one is carried out after the ignition is turned to the RUN position and the other is carried out after an ABS stop or after an AdvanceTrac® system activation. When the ignition is turned to the RUN position, the ABS module carries out a preliminary electrical check of the wheel speed sensors and their circuits by sending voltage through the sensor and checking for the voltage to return. When the vehicle is traveling at speeds above 21 km/h (13 mph) and the brake pedal is not being pressed, the HCU pump motor is commanded on for approximately 0.5 second to check pump motor operation. Also, during all phases of operation while the vehicle is in motion, the ABS module checks for correct operation of the wheel speed sensors by comparing wheel speed input to other sensor input. Depending on the nature of the malfunction detected, the ABS module deactivates the anti-lock control system and sends a message over the HS-CAN to the IPC . When the IPC receives this message, it illuminates the yellow ABS warning indicator and the stability/traction control indicator (sliding car icon).

Electronic Brake Distribution (EBD)

On initial application of the brake pedal, full pressure is applied to the rear brakes. The ABS module then uses wheel speed input to calculate an estimated rate of deceleration. Once vehicle deceleration exceeds a predetermined threshold, the ABS module commands the **HCU** to close the appropriate isolation valves to hold the rear brake pressure constant while allowing the front brake pressure to build. This creates a balanced braking condition between the front and rear wheels and minimizes the chance of rear wheel lockup during hard braking. As the vehicle decelerates, the valves are opened to increase the rear brake pressure in proportion to the front brake pressure.

If ABS is disabled due to DTCs being present in the ABS module, **EBD** continues to function unless the DTCs are for wheel speed sensors or the **HCU**. When **EBD** is disabled, the red brake warning light, the ABS warning indicator and the stability/traction control indicator (sliding car icon) illuminate.

Supplemental Braking Assist

The ABS module uses the **HCU** and hydraulic pump motor to aid in bringing the vehicle to a safe, controlled stop in the event of severe vacuum loss at the brake booster. The ABS module continually monitors the vacuum in the brake booster through the use of a vacuum sensor. When the vacuum sensor indicates vacuum is below a predetermined level, a DTC is set in the ABS module. If this occurs during a braking event or if the driver attempts to stop the vehicle with a low vacuum condition in the brake booster, the ABS module activates the hydraulic pump motor in the **HCU** to assist with vehicle braking.

Traction Control Function

The ABS module continuously monitors and compares the rotational speed of the drive wheels in relation to the non-driven wheels. When the drive wheels begin to spin faster than the non-driven wheels, the ABS module commands the **HCU** to modulate brake pressure to the appropriate brake caliper. This is accomplished by opening and closing the appropriate solenoid valves inside of the **HCU** while the hydraulic pump motor is activated. At the same time, the ABS module sends a message over the **HS-CAN** that a traction control event is taking place. When the **IPC** receives this message, it flashes the stability/traction control indicator (sliding car icon). If the event is severe enough, the ABS module sends a message to the PCM over the **HS-CAN** to assist with traction control. When the PCM receives this message, it adjusts engine timing and decreases fuel injector pulses. Once the affected wheel returns to normal speed, the ABS module returns the solenoid valves in the **HCU** to their normal position, deactivates the hydraulic pump motor and sends another message over the **HS-CAN** indicating that the traction event has ended. The PCM returns engine timing and fuel injectors to normal operation and the **IPC** turns off the sliding car icon.

Once vehicle speed reaches or exceeds 144 km/h (89 mph), traction control is accomplished only through PCM intervention.

Certain DTCs in the ABS module disable the traction control system. Depending on the DTCs present, the red brake warning light, the yellow ABS warning indicator and the sliding car icon illuminate.

Electronic Stability Control (ESC) and Roll Stability Control (RSC®) Functions

The **ESC** and **RSC®** functions are controlled by the ABS module and use the same wheel speed sensors and tone rings used for anti-lock braking. The **ESC** and **RSC®** functions also use input from the steering wheel rotation sensor, the stability control sensors (yaw rate, roll rate, longitudinal and lateral accelerometers) which are internal to the **RCM** and information from other modules sent over the **HS-CAN** to help maintain vehicle stability.

The ABS module uses the various sensors to continuously monitor vehicle roll motion and direction of travel relative to the driver's intended course. If the ABS module determines from all these inputs that the vehicle is unable to travel in the intended direction or that a condition exists for a potential rollover event, it modulates brake pressure to the appropriate brake caliper(s) by opening and closing the appropriate solenoid valves inside the **HCU** while the hydraulic pump motor is activated. At the same time, the ABS module sends a message over the **HS-CAN** that a stability event is taking place. When the **IPC** receives this message, it flashes the stability/traction control indicator (sliding car icon). During a stability event, conditions may require the involvement of the PCM to limit engine torque like it does for traction control. Once the stability condition has been corrected, the ABS module sends another message over the **HS-CAN** indicating that the event has ended. The PCM returns engine timing and fuel injectors to normal operation and the **IPC** turns off the sliding car icon.

Certain DTCs being present in the ABS module may disable the **ESC** and **RSC®** functions. Depending on the DTCs present, the yellow ABS warning indicator and/or the sliding car icon are illuminated.

Trailer Sway Control Function

Trailer sway control is a unique function of the **RSC®** system that uses information from the steering wheel rotation sensor and the yaw information from the **RCM** to determine if a trailer sway event is taking place. Trailer sway is the undesirable yaw force a trailer can apply to the towing vehicle. If the ABS module determines that a trailer sway event is taking place, it takes action similar to the action taken for an **ESC** and an **RSC®** event. At the same time, the ABS module sends a message over the **HS-CAN** that a trailer sway event is taking place. When the **IPC** receives this message, it flashes the stability traction control indicator (sliding car icon) and displays TRAILER SWAY REDUCE SPEED in the message center. If necessary, the PCM reduces engine torque as it does for **ESC** and **RSC®**. Once the trailer sway has been corrected, the ABS module sends another message over the **HS-CAN** indicating that the event has ended. The PCM returns engine timing and fuel injectors to normal operation and the **IPC** turns off the stability traction control indicator (sliding car icon).

If the vehicle is also equipped with a **TBC** system, the **TBC** module receives the trailer sway message from the ABS module and takes appropriate action to assist in reducing the trailer sway. Refer to Section 206-10 for information on the **TBC** system.

Trailer sway control only activates with vehicle speed greater than 64 km/h (40 mph). Any malfunction that disables the **RSC®** function also disables the trailer sway control function. Trailer sway control can also be disabled through the message center. Refer to the Owner's Literature for the disabling/enabling procedure. Regardless of the chosen state (enabled or disabled), trailer sway control is re-enabled at each ignition key cycle.

Hill Start Assist

The hill start assist system engages automatically when the following conditions are met:

- The driver applies the vehicle brake and the vehicle is at a complete stop.
- Road inclination of approximately 4 degrees or more is detected (as indicated by the stability control sensor information).
- The vehicle transmission information indicates that the driver selects the correct gear, indicating the intention to drive up the hill.
- The parking brake is not engaged.

Once all the above criteria have been met and the driver releases the brake pedal, hill start assist utilizes the valves in the Hydraulic Control Unit (HCU) to hold the car stationary for up to 2 seconds giving the driver time to accelerate the vehicle. As the vehicle accelerates hill start assist gradually releases the brake pressure to make sure that the vehicle is neither rolling back nor driving off until there is sufficient driving torque to accelerate the vehicle forward. Hill start assist also functions if the vehicle is facing downhill if the reverse gear is selected.

Hill Descent Control™ and Hill Descent Control™ Switch

Hill Descent Control™ uses the pump motor and **HCU** to provide vehicle speed control while descending a downhill grade, information from the **RCM** to determine whether or not the vehicle is on a downhill grade and wheel speed sensor information to determine vehicle speed. When the Hill Descent Control™ switch is pressed, a ground signal is sent to the **IPC** indicating that the driver has activated the system. The **IPC** then sends a message to the ABS module along the **HS-CAN** requesting system activation. The ABS module uses the input from the **RCM** and wheel speed sensors to determine the mode of operation (active, stand-by or deactivated). During a hill descent, while the Hill Descent Control™ system is activated, the driver sets vehicle speed through the use of the accelerator and brake pedals. Once the desired speed has been set, the system maintains that speed until one of the following occur:

- The vehicle is no longer on a downhill grade.
- The driver presses the switch again.
- Vehicle speed exceeds 64 km/h (40 mph).
- The system is disabled for cooling.

During periods of sustained use, the Hill Descent Control™ system automatically deactivates in order to cool the brake system. When this occurs, the system notifies the driver with a message in the message center and by sounding a warning chime. At this time the driver needs to provide brake and/or throttle control to maintain the desired descent speed. The system remains disabled until it has sufficiently cooled. The length of time needed for cooling varies depending on operating conditions.

Stability/Traction Control Switch

Unlike conventional ABS, the individual functions of AdvanceTrac® with RSC® can be deactivated by the driver through the stability/traction control switch. The switch is hard-wired to the IPC and when pressed, sends a ground signal to the IPC.

If the stability/traction control switch is pressed and held for more than 30 seconds, a DTC sets in the ABS module.

Once disabled, the system remains in that state until the driver presses the switch again, the ignition is cycled or the vehicle speed reaches or exceeds 56 km/h (35 mph). Once the vehicle speed reaches or exceeds 56 km/h (35 mph), RSC® and ESC are reactivated. Once the vehicle speed drops below 48 km/h (30 mph), traction control returns to the state the driver had originally selected, either on or off.

The functions cannot be disabled when a MyKey® restricted key is in use with the AdvanceTrac® always-on feature configured to on. If an attempt is made to disable the ESC system while a MyKey® restricted key is in use, ADVTRAC ON MY KEY SETTING displays in the IPC message center. For information about MyKey®, refer to Section 419-01B.

Stability/Traction Control Indicators

Both the stability/traction control indicator (sliding car icon) and the stability/traction control disabled indicator (sliding car OFF icon) alert the driver of stability/traction control system status.

When certain stability control system DTC(s) are present, the ABS module sends a message to the IPC over the HS-CAN. Depending on the nature of the DTC(s), the IPC responds by either solidly illuminating 1 or both icons or by flashing the icon(s) once every 2 seconds.

When a stability/traction control event is taking place, the ABS module sends a message to the IPC over the HS-CAN to flash the sliding car icon twice per second.

When the driver disables the stability/traction control system by using the traction control switch, the IPC receives a ground signal from the switch and then sends a message over the HS-CAN indicating that the driver has requested the system be deactivated and illuminates the sliding car OFF icon.

Stability Control Sensors

The stability control sensors provide information to the ABS module for the operation of the ESC function, RSC® function (if equipped), hill start assist system and the Hill Descent Control™ system (if equipped). The stability control sensors are housed in the RCM and are serviced with the module. The sensors used for stability control include the yaw rate sensor, the lateral accelerometer, the longitudinal accelerometer and (if equipped with RSC®) the roll rate sensor. Sensor information is sent to the ABS module over a private HS-CAN.

Steering Wheel Sensor Information

For the 6.2L engine, the steering wheel rotation sensor uses an optical sensor and a slotted wheel to measure the rate of rotation (angle) of the steering wheel. The steering wheel rotation sensor uses the HS-CAN to transmit information to the ABS module about steering wheel turning direction (left or right) and how far the steering wheel is being turned. The steering wheel rotation sensor does not indicate the absolute position of the steering wheel relative to straight-ahead. The ABS module learns this position by comparing the steering wheel position with other signals and storing the position it has learned. The module confirms this position and modifies it as necessary during every new driving cycle.

For all other applications, the steering wheel rotation sensor is part of the EPAS system and is located inside the PSCM. The sensor is serviced with the PSCM. The PSCM uses the HS-CAN to transmit information to the ABS module about steering wheel turning direction (left or right) and how far it is being turned. The PSCM does not indicate the absolute position of the steering wheel relative to straight-ahead. The ABS module learns this position by comparing the steering wheel position with other signals and storing the position it has learned. The ABS module confirms this position and modifies it as necessary during every new driving cycle.

ABS Module Configuration

Module configuration is a scan tool process which configures specific vehicle settings in a new module. If installing a new ABS module and/or a new HCU , the ABS module must be configured. To configure the ABS module, refer to PMI in Section 418-01.

Inspection and Verification

1. Verify the customer concern.
2. Verify the stoplamps operate correctly by applying and releasing the brake pedal. If the stoplamps do not operate correctly, refer to Section 417-01. If the stoplamps operate correctly, proceed to the next step.
3. Verify that the PRNDL operates correctly on the IPC . If the PRNDL does not operate correctly, refer to Section 413-01. If the PRNDL operates correctly, proceed to the next step.
4. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Wheel speed sensor ring • HCU • Tire pressure • Mismatched wheels or tires on vehicle • Base brake system • Brake booster vacuum hose • Steering components, suspension components and tire size	• BJB fuses: ▪ 36 (30A) ▪ 47 (60A) ▪ 54 (5A) • ABS module • Wheel speed sensors • Stoplamp switch • Stability/traction control switch • Hill Descent Control™ switch (if equipped) • Brake fluid level switch • Wiring, terminals or connectors • Brake booster vacuum sensor

5. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
6. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the DLC .

7. **NOTE:** *The VCM LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

8. If the scan tool does not communicate with the vehicle:
 - verify the ignition is ON.
 - verify the scan tool operation with a known good vehicle.
 - refer to Section 418-00 The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

9. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record **CMDTCs** .
10. Clear the continuous DTCs and carry out the self-test diagnostics for the ABS module.
11. If the DTCs retrieved are related to the concern, go to ABS Module DTC Chart. For all other DTCs, refer to the Master DTC Chart in Section 419-10.
12. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

ABS Module DTC Chart

NOTE: Always retrieve the continuous DTCs after running the self-test.

DTC	Description	Action
B1342	ECU is Defective	NOTICE: Do not power the Electronic Control Unit (ECU) while it is removed from the HCU . Failure to do so could damage the ECU or cause the ECU to set unrelated DTCs. NOTE: If other DTCs are present, repair them before installing a new component. CLEAR the DTCs. REPEAT the self-test. RETRIEVE the DTCs. If DTC B1342 is retrieved, INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. REPEAT the self-test.
B1596	Service Continuous Codes	NOTE: This DTC does not indicate a malfunction of the ABS. If other DTCs are present, REPAIR them before addressing this DTC. Some DTCs cannot be detected by the ABS module until the ignition is cycled from ON to OFF and back to ON again. When DTCs are cleared using the scan tool, the ignition needs to be cycled in order to clear this DTC and have the ABS module check for other DTCs. DIAGNOSE all other DTCs. CLEAR the DTCs. CYCLE the ignition. RETRIEVE the DTCs. GO to the ABS Module DTC Chart.
B1676	Battery Pack Voltage Out of Range	GO to Pinpoint Test A.
B2477	Module Configuration Failure	CONFIGURE the ABS module. REFER to Programmable Module Installation (PMI) in Section 418-01. CLEAR the DTCs. REPEAT the self-test.
B2900	VIN Mismatch	GO to Pinpoint Test B.
C109D	Low Vacuum Condition Detected	GO to Pinpoint Test I.
C109E	Vacuum Sensor Circuit Fault	GO to Pinpoint Test I.
C109F	Vacuum Sensor Signal Fault	GO to Pinpoint Test I.
C1145	Wheel Speed Sensor RF Input Circuit Failure	GO to Pinpoint Test D.
C1155	Wheel Speed Sensor LF Input Circuit Failure	GO to Pinpoint Test D.

DTC	Description	Action
C1165	Wheel Speed Sensor RR Input Circuit Failure	GO to Pinpoint Test D.
C1175	Wheel Speed Sensor LR Input Circuit Failure	GO to Pinpoint Test D.
C116A	ABS Pressure Transducer / Brake Switch Mismatch	GO to Pinpoint Test G.
C1222	Wheel Speed Mismatch	GO to Pinpoint Test E.
C1266	ABS Valve Power Relay Circuit Failure	GO to Pinpoint Test F.
C1278	Steering Wheel Angle 1 and 2 Signal Faulted	GO to Pinpoint Test U.
C1280	Yaw Rate Sensor Signal Fault	This DTC indicates a concern with the yaw rate sensor which is integral to the Restraints Control Module (RCM) . REFER to Section 501-20B to diagnose all the RCM DTCs. If no RCM DTCs are present, GO to Pinpoint Test P.
C1282	Lateral Accelerometer Signal Fault	This DTC indicates a concern with the lateral accelerometer which is integral to the RCM . REFER to Section 501-20B to diagnose all the RCM DTCs. If no RCM DTCs are present, GO to Pinpoint Test P.
C1288	Pressure Transducer Main/Primary Input Circuit Failure	CLEAR the DTCs. REPEAT the self-test. RETRIEVE the DTCs. If DTC C1288 is retrieved, INSTALL a new Hydraulic Control Unit (HCU) . REFER to Hydraulic Control Unit (HCU). CLEAR the DTCs. REPEAT the self-test.
C1296	Wheel Speed LF Signal Fault	GO to Pinpoint Test E.
C1297	Wheel Speed RF Signal Fault	GO to Pinpoint Test E.
C1298	Wheel Speed RR Signal Fault	GO to Pinpoint Test E.
C1299	Wheel Speed LR Signal Fault	GO to Pinpoint Test E.
C1300	ABS Pump Motor Circuit Failure	GO to Pinpoint Test C.
C1440	Pressure Transducer Main/Primary Signal Faulted	GO to Pinpoint Test G.
C1516	Roll Rate Sensor Signal Fault	This DTC indicates a concern with the roll rate sensor which is integral to the RCM . REFER to Section 501-20B to diagnose all the RCM DTCs. If no RCM DTCs are present, GO to Pinpoint Test P.
C1963	Stability Control Inhibit Warning	GO to Pinpoint Test P.
C1975	IVD Plausibility Failure	GO to Pinpoint Test P.
C2770	Longitudinal Acceleration Sensor Plausibility Failure	This DTC indicates a concern with the longitudinal accelerometer which is integral to the RCM . REFER to Section 501-20B to diagnose all the RCM DTCs. If no RCM DTCs are present, GO to Pinpoint Test P.
C2775	TC/AYC ON/OFF Switch Stuck Failure	GO to Pinpoint Test R.
C2809	Hill Descent Control Switch Stuck	GO to Pinpoint Test J.

DTC	Description	Action
U0073	Control Module Communication Bus A Off	A HS-CAN fault was present at a point in time. The fault is not currently present since the module is communicating with the diagnostic scan tool. CLEAR the DTC. REPEAT the Network Test with the diagnostic scan tool. VERIFY the integrity of the connectors and wiring. Refer to Wiring Diagrams Cell 14 for schematic and connector information.
U0100	Loss of Communication with ECM/PCM	GO to Pinpoint Test K.
U0114	Lost Communication With Four-Wheel Drive Clutch Control Module	GO to Pinpoint Test L.
U0131	Lost Communication With Power Steering Control Module (VAPS)	GO to Pinpoint Test M. For the 6.2L engine, GO to Pinpoint Test N.
U0140	Lost Communication With Body Control Module (GEM)	GO to Pinpoint Test O.
U0151	Lost Communication With Restraints Control Module	GO to Pinpoint Test T.
U0155	Lost Communication With Instrument Panel Cluster (IC) Control Module	GO to Pinpoint Test Q.
U0401	Invalid Data Received From ECM/PCM	This DTC sets when the ABS module receives invalid network data from the PCM. RETRIEVE and REPAIR all non-network DTCs in the PCM. REFER to Section 303-14.
U0414	Invalid Data Received from Four-Wheel Drive Clutch Control Module	This DTC sets when the ABS module receives invalid network data from the Transfer Case Control Module (TCCM) . RETRIEVE and REPAIR all non-network DTCs in the TCCM . REFER to Section 308-07A.
U0423	Invalid Data Received From Instrument Panel Control Module	This DTC sets when the ABS module receives invalid network data from the Instrument Panel Cluster (IPC) . RETRIEVE and REPAIR all non-network DTCs in the IPC . REFER to Section 413-01.
U0428	Invalid Data Received from Steering Angle Sensor Module	This DTC sets when the ABS module receives invalid steering angle network data. For vehicles equipped with Hydraulic Power Assist Steering (HPAS) , this data is sent from the Steering Column Control Module (SCCM) . RETRIEVE and REPAIR all non-network DTCs in the SCCM . REFER to Section 211-05. For vehicles equipped with Electronic Power Assist Steering (EPAS) , this data is sent from the Power Steering Control Module (PSCM) . RETRIEVE and REPAIR all non-network DTCs in the PSCM . REFER to Section 211-02.
U0452	Invalid Data Received From Restraints Control Module	This DTC sets when the ABS module receives invalid network data from the RCM . RETRIEVE and REPAIR all non-network DTCs in the RCM . REFER to Section 501-20B. If no RCM DTCs are present, GO to Pinpoint Test T.
U2050	No Application Present	CONFIGURE the ABS module. REFER to Programmable Module Installation (PMI) in Section 418-01. CLEAR the DTCs. REPEAT the self test.
U2051	One or More Calibration Files Missing /Corrupt	CLEAR the DTC. CARRY OUT the ABS module self-test. If DTC U2051 returns, RECONFIGURE the ABS module using as-built data. REFER to PMI Using the Integrated Diagnostic System (IDS) When the Original Module is NOT Available in Section 418-01. CLEAR the DTC and REPEAT the self-test. If DTC U2051 returns, INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. CLEAR the DTC. REPEAT the self-test.

DTC	Description	Action
U2152	SCP (J1850) Invalid Data from GEM	This DTC sets when the ABS module receives invalid network data from the Body Control Module (BCM) . RETRIEVE and REPAIR all non-network DTCs in the BCM . REFER to Section 419-10.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
No communication with the ABS module	- Fuse - Wiring, terminals or connectors - ABS module	REFER to Section 418-00.
The yellow ABS warning indicator is never on	- Wiring, terminals or connectors - Instrument Panel Cluster (IPC)	REFER to Section 413-01.
The stability/traction control indicator (sliding car icon) is never on	- Stability/traction control system - ABS module - IPC	REFER to Section 413-01.
Only the red brake warning indicator is always/never on	- Wiring, terminals or connectors - Parking brake switch - Brake fluid level switch - Body Control Module (BCM) - IPC	REFER to Section 413-01.
The yellow ABS warning indicator is always on	- DTCs in the ABS module - ABS module - IPC	- RETRIEVE and RECORD any ABS module DTCs. GO to the ABS Module DTC Chart. - If there are no DTCs present in the ABS module, REFER to Section 413-01.
The sliding car icon is always on	- DTCs in the ABS module - Stability/traction control system - IPC	- RETRIEVE and RECORD any ABS module DTCs. GO to the ABS Module DTC Chart. - If there are no DTCs present in the ABS module, REFER to Section 413-01.
All 3 indicators (red brake warning light, yellow ABS warning	- DTCs in the ABS module - Wiring, terminals or connectors	RETRIEVE and RECORD any ABS module DTCs. GO to the ABS Module DTC Chart.

indicator and the sliding car icon) are always on	- Parking brake switch - Brake fluid level switch - Stability/traction control system - BCM - ABS module - IPC	If there are no DTCs present in the ABS module, REFER to Section 413-01.
Spongy brake pedal with no warning indicator	- Air in the brake hydraulic system - Damaged or incorrectly installed brake components	- BLEED the brake system and the Hydraulic Control Unit (HCU) . REFER to Brake System Bleeding in Section 206-00. - INSPECT the brake pedal, brake booster and master cylinder for correct installation. REFER to Section 206-06 for the brake pedal and master cylinder or Section 206-07 for the brake booster. CORRECT the installation or INSTALL new as necessary. TEST the system for normal operation.
Poor vehicle tracking during anti-lock function	- Air in the hydraulic brake system - Low tire pressure - HCU - Base brake system concern	GO to Pinpoint Test S. - REFER to Section 206-00.
The stability/traction control system is inoperative	- Wiring, terminals or connectors - Stability/traction control switch - ABS module - IPC	GO to Pinpoint Test R.
The stability/traction control system cannot be disabled	- If a MyKey® restricted key is in use with the AdvanceTrac® always on feature configured to on, the traction control system cannot be disabled. An admin key must be used in order to enable and disable the traction control system - Wiring, terminals or connectors - Stability/traction control switch	GO to Pinpoint Test R.

	• IPC	
• The Hill Descent Control™ system (if equipped) is inoperative	• ABS module DTCs present • Hill Descent Control™ switch • Wiring, terminals or connectors	• CARRY OUT a ABS module self-test. • GO to Pinpoint Test J.

Pinpoint Tests

Pinpoint Test A: DTC B1676

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The voltage required for correct ABS operation is between 10 and 17 volts. Voltage outside of this range causes the ABS module to set DTC B1676.

- DTC B1676 (Battery Pack Voltage Out of Range) — this DTC sets in every ignition cycle if, with the ignition in the RUN position and vehicle speed greater than 6 km/h (4 mph), the battery voltage is less than 10 volts or more than 17 volts. DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected.

This pinpoint test is intended to diagnose the following:

- Fuse
- Battery condition
- Wiring, terminals or connectors
- ABS module

PINPOINT TEST A : DTC B1676

NOTE: *DTC B1676 can be set if the vehicle has been recently jump started, the battery has been recently charged or the battery has been discharged. The battery may become discharged due to excessive load(s) on the charging system from aftermarket accessories or if the battery has been left unattended with the accessories on.*

NOTE: *Carry out a thorough inspection and verification before proceeding with the pinpoint test. Refer to Inspection and Verification in this section.*

A1 RETRIEVE ALL CMDTC IN ALL MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — All CMDTCs .

Is DTC B1317, B1318 or B1676 present in one or more modules AND P0563, P0620, P0625, P0626 or P065B present in the PCM?

Yes	REFER to Section 414-00 for diagnosis of the battery and charging system.
No	GO to A2 .

A2 CHECK BATTERY CONDITION

- Ignition OFF.
- Carry out the Battery — Condition Test. Refer to Section 414-01.

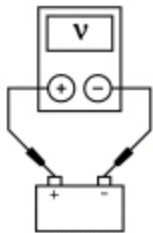
Does the battery pass the condition test?

Yes	If the battery passed the condition test but required a recharge, REFER to Section 414-00 to diagnose the charging system. CLEAR all <u>CMDTCs</u> . TEST the system for normal operation. If the battery passed the condition test and did not require a recharge, GO to A3 .
No	INSTALL a new battery. REFER to Section 414-01. CLEAR all <u>CMDTCs</u> . TEST the system for normal operation.

A3 CHECK CHARGING SYSTEM VOLTAGE

NOTE: Do not allow the engine speed to increase above 2,000 rpm while carrying out this step or the generator can self excite and result in default charging system output voltage. If engine speed goes above 2,000 rpm, shut the vehicle off and restart the engine before carrying out this step.

- Start the engine.
- Measure the voltage of the battery with and without a load on the charging system as follows:
 - Turn off all accessories and run the engine at 1,500 rpm for a minimum of 2 minutes while measuring battery voltage.
 - Turn the headlights on and the HVAC fan on high and run engine at 1,500 rpm for a minimum of 2 minutes while measuring battery voltage.



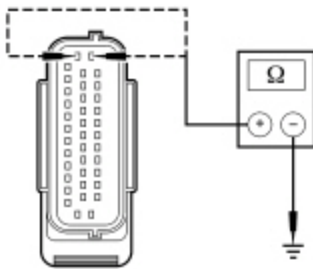
AJ0210-A

Are the voltages between 13 and 15.2 volts?

Yes	GO to A4 .
No	REFER to Section 414-00 to diagnose the charging system. CLEAR all <u>CMDTCs</u> . TEST the system for normal operation.

A4 CHECK THE ABS MODULE GROUND CIRCUITS FOR AN OPEN

- Disconnect: Battery Negative Cable .
- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Measure the resistance between ground and:
 - ABS module [C135](#) Pin 13, circuit GD121 (BK/YE), harness side.
 - ABS module [C135](#) Pin 38, circuit GD121 (BK/YE), harness side.



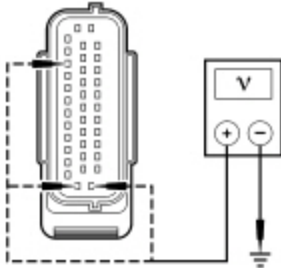
N0055395

Are the resistances less than 0.5 ohm?

Yes	GO to A5 .
No	REPAIR the affected circuit(s). CLEAR all <u>CMDTCs</u> . REPEAT the self-test.

A5 CHECK FOR VOLTAGE AT THE ABS MODULE

- Connect: Battery Negative Cable .
- Ignition ON.
- Measure the voltage between ground and:
 - ABS module [C135](#) Pin 1, circuit SBB47 (WH/RD), harness side.
 - ABS module [C135](#) Pin 25, circuit SBB36 (GN/RD), harness side.
 - ABS module [C135](#) Pin 35, circuit CBB54 (VT/OG), harness side.



N0089362

Are the voltages greater than 10 volts?

Yes	GO to A6 .
No	VERIFY that Battery Junction Box (BJB) fuses 36 (30A), 47 (60A), and 54 (5A) are OK. If OK, REPAIR the affected circuit(s). If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the self-test.

A6 CHECK FOR CORRECT ABS MODULE OPERATION

- Disconnect: ABS Module [C135](#).
- Check the connector for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: ABS Module [C135](#).

- Make sure the connector seats correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test B: DTC B2900

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

When the ignition is turned to the RUN position, the ABS module and the Body Control Module (BCM) share Vehicle Identification Number (VIN) information over the High Speed Controller Area Network (HS-CAN) .

- DTC B2900 (VIN Mismatch) — this DTC sets in the ignition cycle if the VIN information stored in the ABS module does not match the information transmitted by the BCM . DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected.

This pinpoint test is intended to diagnose the following:

- Module configuration (ABS and BCM)
- ABS module

PINPOINT TEST B : DTC B2900

B1 VERIFY BCM VIN	
• Connect the scan tool. • Ignition ON. • Using the scan tool, carry out the network test. • Enter the following diagnostic mode on the scan tool: Log Viewer . • Compare the VIN in log viewer to the VIN plate.	
Does VIN in log viewer match the VIN plate?	
Yes	GO to B2 .
No	RECONFIGURE the BCM . FOLLOW the instructions on the scan tool. CLEAR the DTCs. CYCLE the ignition key. REPEAT the self-test.
B2 VERIFY ABS MODULE PART NUMBER	
• Retrieve and record the ABS module part number from log viewer and verify that the vehicle has the correct ABS module installed.	

Is the correct ABS module installed in the vehicle?

Yes	CONFIGURE the ABS module. REFER to Programmable Module Installation (PMI) Using the Integrated Diagnostic System (IDS) When the Original Module is NOT Available in Section 418-01. CLEAR the DTC. CYCLE the ignition key. REPEAT the self-test.
No	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. CLEAR the DTC. CYCLE the ignition key. TEST the system for normal operation.

Pinpoint Test C: DTC C1300

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The operating voltage required to run the hydraulic pump motor is between 10 and 17 volts. Fused battery voltage for the hydraulic pump motor is supplied to the ABS module from the Battery Junction Box (BJB) . The hydraulic pump motor is internal to the Hydraulic Control Unit (HCU) . If a new hydraulic pump motor is needed, a new HCU must be installed.

- DTC C1300 (ABS Pump Motor Circuit Failure) — this DTC sets every ignition cycle or in on-demand memory if there are circuit failures such as a short to ground, open, high circuit resistance, damaged motor, blown fuse. The module also checks the motor after every actuation. When the pump is shut off, the ABS module measures the output voltage of the motor. If the voltage drops too quickly, there may be a locked motor. DTC C1300 sets if this fault is detected 3 times. DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- HCU
- ABS module

PINPOINT TEST C : DTC C1300

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
C1 CHECK THE ABS PUMP MOTOR	
• Ignition ON.	
Is the ABS pump motor running all the time?	
Yes	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.
No	GO to C2 .
C2 CHECK THE ABS MODULE PUMP MOTOR (PMP_MOTOR) OUTPUT COMMAND	
• Connect the scan tool. • Ignition ON.	

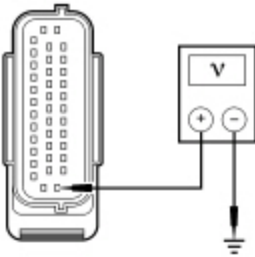
- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- Toggle the PMP_MOTOR output command ON.

Does the ABS pump motor run for approximately 3 seconds?

Yes	TOGGLE the PMP_MOTOR output command OFF. CLEAR the DTCs. CHECK the yellow ABS warning indicator while driving the vehicle (the brakes must not be applied) above 32 km/h (20 mph). If the yellow ABS warning indicator illuminates, RETRIEVE the DTCs. If DTC C1300 is retrieved again, GO to C5 . If the yellow ABS warning indicator does not illuminate, the system is OK.
No	TOGGLE the PMP_MOTOR output command OFF. GO to C3 .

C3 CHECK FOR VOLTAGE TO THE ABS MODULE

- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Ignition ON.
- Measure the voltage between ABS module [C135](#) Pin 1, circuit SBB47 (WH/RD), harness side and ground.



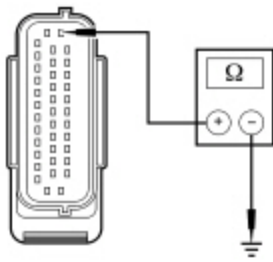
• N0058091

Is the voltage greater than 10 volts?

Yes	GO to C4 .
No	VERIFY BJB fuse 47 (60A) is OK. If OK, REPAIR open in circuit SBB47 (WH/RD). If not OK, REPAIR the short circuit in circuit SBB47 (WH/RD). CLEAR the DTCs. REPEAT the self-test.

C4 CHECK THE ABS MODULE GROUND CIRCUIT FOR AN OPEN OR EXCESSIVE RESISTANCE

- Ignition OFF.
- Disconnect: Battery Negative Cable .
- Measure the resistance between ABS module [C135](#) Pin 13, circuit GD121 (BK/YE), harness side and ground.



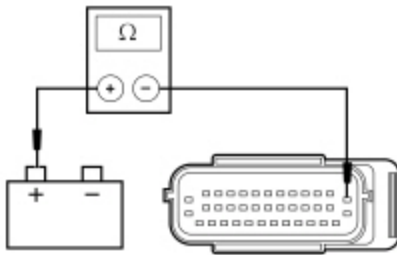
• N0055398

Is the resistance less than 0.5 ohm?

Yes	GO to C5 .
No	REPAIR circuit GD121 (BK/YE). CLEAR the DTCs. REPEAT the self-test.

C5 CHECK THE ABS MODULE POWER CIRCUIT FOR CONTINUITY

- Measure the resistance between the ABS module [C135](#) Pin 1, circuit SBB47 (WH-RD), harness side and battery positive.



• N0117697

Is the resistance less than 0.5 ohms?

Yes	GO to C6 .
No	REPAIR the open circuit SBB47 (WH/RD). CLEAR the DTCs. REPEAT the self-test.

C6 CHECK FOR CORRECT ABS MODULE OPERATION

- Disconnect: ABS Module [C135](#).
- Check the connector for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: ABS Module [C135](#).
- Connect: Battery Negative Cable .
- Make sure the connector seats correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new HCU . REFER to Hydraulic Control Unit (HCU) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test D: DTCs C1145, C1155, C1165 and C1175

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

Active wheel speed sensors generate a square wave current signal that is sent to the ABS module. The wheel speed sensor circuitry connects to the ABS module through 2 wires and a connector at each wheel speed sensor. When the ignition switch is turned to the RUN position, the ABS module carries out a self-test by sending a reference voltage through the wheel speed sensors and their circuitry to determine if they are functional. The front wheel bearings contain a magnetic strip that replaces the tone ring. It functions the same but is not visible for inspection.

Battery voltage and ground are supplied to the active wheel speed sensors from the ABS module.

DTC Description	Fault Trigger Condition
• C1145 — Speed Wheel Sensor RF Input Circuit Failure • C1155 — Speed Wheel Sensor LF Input Circuit Failure • C1165 — Speed Wheel Sensor RR Input Circuit Failure • C1175 — Speed Wheel Sensor LR Input Circuit Failure	These DTCs set in continuous memory if the ABS module detects an open, short to ground or voltage or a damaged wheel speed sensor input circuit, the appropriate DTC sets when the vehicle speed exceeds 6 km/h (4 mph). DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected. DTCs C1165 and C1175 also set on-demand.

This pinpoint test is intended to diagnose the following:

- Wheel speed sensors
- Wheel bearing magnetic strips, rear tone rings
- Wiring, terminals or connectors
- ABS module

PINPOINT TEST D : DTCS C1145, C1155, C1165 AND C1175

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *If the ignition is turned to the ON position when a wheel speed sensor is disconnected, the ABS module disables the output to that sensor. Make sure all test connections are made at the wheel speed sensor connector before turning the ignition to the ON position.*

D1 CHECK FOR FAULT REPEATABILITY

- Connect the scan tool.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear the Continuous DTCs .
- Drive the vehicle at least 16 km/h (10 mph).
- Retrieve the Continuous Memory Diagnostic Trouble Codes (CMDTCs) .

Is DTC C1145, C1155, C1165 or C1175 retrieved?

Yes	If the Rotunda Active Wheel Speed Sensor Tester is available, GO to D2 . If the Rotunda Active Wheel Speed Sensor Tester is not available, GO to D4 .
No	INSPECT the wheel speed sensors and wheel speed sensor wiring. REPAIR or INSTALL new as necessary. If any other DTCs are retrieved, GO to the ABS Module DTC Chart.

D2 CHECK THE ABS MODULE OUTPUT USING THE ROTUNDA ACTIVE WHEEL SPEED SENSOR TESTER

- Ignition OFF.
- Disconnect: Suspect Wheel Speed Sensor .
- Connect the Rotunda Active Wheel Speed Sensor Tester to the wheel speed sensor connectors.
- Select the correct system polarity on the tester and turn the tester power switch to the ON position.
- Ignition ON.

Is the module output LED illuminated?

Yes	GO to D3 .
No	GO to D6 .

D3 CHECK THE WHEEL SPEED SENSOR OUTPUT WITH THE ROTUNDA ACTIVE WHEEL SPEED SENSOR TESTER

- Raise the suspect wheel until it can spin freely. Refer to Section 100-02.
- While monitoring the Rotunda Active Wheel Speed Sensor Tester, slowly spin the suspect wheel.

Do the sensor output LEDs illuminate and flash and is the current overload LED not illuminated?

Yes	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

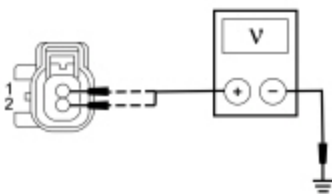
No	If the current overload level LED is not illuminated and the sensor output LEDs do not illuminate or if the current overload level LED is illuminated red, INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor — Front or Wheel Speed Sensor — Rear in this section. CLEAR the DTCs. REPEAT the self-test. If the current overload level LED is not illuminated and the sensor output LEDs illuminate green but do not flash, INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor — Front or Wheel Speed Sensor — Rear in this section. CLEAR the DTCs. REPEAT the self-test. If the outputs LEDs still do not flash when spinning the wheel, INSTALL a new wheel bearing magnetic strip or tone ring (rear). REFER to Section 204-01A, Section 204-01B or Wheel Speed Sensor Ring — Rear in this section.
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D4 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR A SHORT TO VOLTAGE

NOTE: Check both circuits for each DTC.

- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Disconnect: Suspect Wheel Speed Sensor .
- Ignition ON.
- Measure the voltage between the suspect wheel speed sensor connector, harness side and ground as follows:

Wheel Speed Sensor	Circuit
C150 Pin 1 (LF)	VCA03 (VT/WH)
C150 Pin 2 (LF)	RCA17 (YE)
C160 Pin 1 (RF)	VCA05 (GY/VT)
C160 Pin 2 (RF)	RCA19 (VT)
C426 Pin 1 (RR)	VCA06 (WH/OG)
C426 Pin 2 (RR)	RCA20 (BN)
C440 Pin 1 (LR)	VCA04 (BU/OG)
C440 Pin 2 (LR)	RCA18 (BN/GN)



N0120698

Is any voltage present?

Yes	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.
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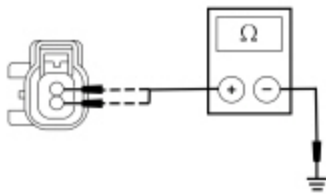
No	GO to D5 .
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D5 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR A SHORT TO GROUND

NOTE: Check both circuits for each DTC.

- Ignition OFF.
- Measure the resistance between the wheel speed sensor connector, harness side and ground as follows:

Wheel Speed Sensor	Circuit
C150 Pin 1 (LF)	VCA03 (VT/WH)
C150 Pin 2 (LF)	RCA17 (YE)
C160 Pin 1 (RF)	VCA05 (GY/VT)
C160 Pin 2 (RF)	RCA19 (VT)
C426 Pin 1 (RR)	VCA06 (WH/OG)
C426 Pin 2 (RR)	RCA20 (BN)
C440 Pin 1 (LR)	VCA04 (BU/OG)
C440 Pin 2 (LR)	RCA18 (BN/GN)



N0120699

Are the resistances greater than 10,000 ohms?

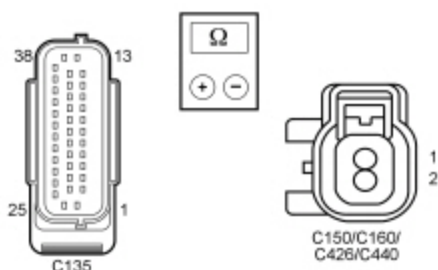
Yes	GO to D6 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.

D6 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR AN OPEN

NOTE: Check both circuits for each DTC.

- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Disconnect: Suspect Wheel Speed Sensor .
- Measure the resistance between the ABS module connector, harness side and the suspect wheel speed sensor connector, harness side as follows:

ABS Module	Circuit	Wheel Speed Sensor
C135 Pin 21	VCA03 (VT/WH)	C150 Pin 1 (LF)
C135 Pin 34	RCA17 (YE)	C150 Pin 2 (LF)
C135 Pin 18	VCA05 (GY/VT)	C160 Pin 1 (RF)
C135 Pin 6	RCA19 (VT)	C160 Pin 2 (RF)
C135 Pin 19	VCA06 (WH/OG)	C426 Pin 1 (RR)
C135 Pin 31	RCA20 (BN)	C426 Pin 2 (RR)
C135 Pin 33	VCA04 (BU/OG)	C440 Pin 1 (LR)
C135 Pin 20	RCA18 (BN/GN)	C440 Pin 2 (LR)



N0122273

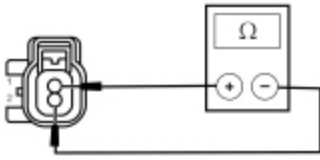
Are the resistances less than 5 ohms?

Yes	GO to D7 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the ABS self-test.

D7 CHECK SHORTED WHEEL SPEED SENSOR CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between the suspect wheel speed sensor connector, harness side as follows:

ABS Module	Circuit	Wheel Speed Sensor
C135 Pin 21	VCA03 (VT/WH)	C150 Pin 1 (LF)
C135 Pin 34	RCA17 (YE)	C150 Pin 2 (LF)
C135 Pin 18	VCA05 (GY/VT)	C160 Pin 1 (RF)
C135 Pin 6	RCA19 (VT)	C160 Pin 2 (RF)
C135 Pin 19	VCA06 (WH/OG)	C426 Pin 1 (RR)
C135 Pin 31	RCA20 (BN)	C426 Pin 2 (RR)
C135 Pin 33	VCA04 (BU/OG)	C440 Pin 1 (LR)
C135 Pin 20	RCA18 (BN/GN)	C440 Pin 2 (LR)



• N0120346

Is the resistance greater than 10,000 ohms?

Yes	GO to D8 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.

D8 CHECK THE ABS MODULE OUTPUT

NOTE: Make sure all test connections are made at the wheel speed sensor before the ignition key is turned to the ON position.

NOTE: Use a meter with a min/max feature and/or recording capabilities to obtain accurate measurements. The ABS module disables the voltage output immediately if the wheel speed sensor is not present.

- Ignition OFF.
- Connect: ABS Module [C135](#).
- Connect the meter to the suspect wheel speed sensor electrical connector pins, harness side.
- Set the meter to measure DC voltage.
- Set the meter range to greater than 15 volts.
- Enable the min/max or recording feature.
- Ignition ON.
- Check the maximum recorded voltage on the meter.

Is the maximum recorded voltage greater than 10 volts?

Yes	INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor — Front or Wheel Speed Sensor — Rear in this section. CLEAR the DTCs. REPEAT the self-test.
No	GO to D9 .

D9 CHECK FOR CORRECT ABS MODULE OPERATION

- Disconnect: ABS Module [C135](#).
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins

- Connect: ABS Module [C135](#).
- Make sure the connector seats correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test E: DTCs C1222, C1296, C1297, C1298 and C1299

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The wheel speed sensor and tone ring generate a square wave current signal to the ABS module that is proportional to the wheel speed. The ABS module compares the wheel speed inputs from all the wheel speed sensors to determine an impending wheel lockup. Incorrect tire size or tire pressure can set these DTCs as well. DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected.

DTC Description	Fault Trigger Condition
• C1222 — Speed Wheel Mismatch	This DTC sets in the ignition cycle when the vehicle speed exceeds 20 km/h (12 mph) or the ABS module detects a difference between wheel speed sensor signals. This DTC can also be set by damaged tone rings, mismatched wheel and/or tire sizes or driving the vehicle on one or more deflated tires.
• C1296 — Wheel Speed LF Signal Fault • C1297 — Wheel Speed RF Signal Fault • C1298 — Wheel Speed RR Signal Fault • C1299 — Wheel Speed LR Signal Fault	These DTCs set in the ignition cycle when the vehicle speed exceeds 20 km/h (12 mph). The appropriate DTC sets if the ABS module does not receive a signal from the wheel speed sensor. These DTCs can also be set by an excessive wheel speed sensor-to-tone ring air gap or a damaged tone ring, mismatched wheel and/or tire sizes or driving the vehicle on one or more deflated tires.

This pinpoint test is intended to diagnose the following:

- ABS module
- Wheel speed sensor magnetic strip/tone ring
- Wheel speed sensor

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: If the ignition is turned to ON when one or more wheel speed sensors are disconnected, the ABS module disables the output to that sensor. Make sure all test connectors are made at the sensor before turning the ignition to ON.

E1 CHECK THE ABS MODULE DTCS

- Review the ABS module DTCS retrieved and recorded during Inspection and Verification.

Is DTC C1145, C1155, C1165 or C1175 also present?

Yes	GO to Pinpoint Test D.
No	GO to E2 .

E2 MONITOR THE WHEEL SPEED SENSOR PIDS

- Connect the scan tool.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- While driving the vehicle at the speed at which the concern occurred, monitor the following PIDs:
 - Left Front Wheel Speed Sensor (LF_WSPD)
 - Right Front Wheel Speed Sensor (RF_WSPD)
 - Left Rear Wheel Speed Sensor (LR_WSPD)
 - Right Rear Wheel Speed Sensor (RR_WSPD)

Are the wheel speed sensor PIDs consistent with the vehicle speed and each other?

Yes	CLEAR the DTCS. DRIVE the vehicle. REPEAT the self-test. RETRIEVE and RECORD any ABS module DTCS. If DTC C1296, C1297, C1298 or C1299 returns, GO to E11. If DTC C1222 is present, GO to E10. For all other ABS module DTCS, GO to the ABS Module DTC Chart. If no DTCS return, INSPECT the ABS module and wheel speed sensor connectors for corrosion, pushed-out pins, or damaged pins and REPAIR if necessary. If all connectors are OK, GO to E7.
No	If the Rotunda Active Wheel Speed Sensor Tester is available, GO to E3. If the Rotunda Active Wheel Speed Sensor Tester is not available, GO to E5.

E3 CHECK THE ABS MODULE OUTPUT USING THE ROTUNDA ACTIVE WHEEL SPEED SENSOR TESTER

- Ignition OFF.
- Disconnect: Suspect Wheel Speed Sensor .
- Connect the Rotunda Active Wheel Speed Sensor Tester to the wheel speed sensor connectors.
- Select the correct system polarity on the Rotunda Active Wheel Speed Sensor Tester and turn the power switch to the ON position.

- Ignition ON.

Is the module output LED illuminated?

Yes	GO to E4 .
No	GO to E6 .

E4 CHECK THE WHEEL SPEED SENSOR OUTPUT WITH THE ROTUNDA ACTIVE WHEEL SPEED SENSOR TESTER

- Raise the suspect wheel until it can spin freely. Refer to Section 100-02.
- While monitoring the Rotunda Active Wheel Speed Sensor Tester, slowly spin the suspect wheel.

Do the sensor output LEDs illuminate and flash and is the current overload LED not illuminated?

Yes	INSPECT the wheel speed sensor tone ring for damage and looseness. INSTALL new components as necessary. If the tone ring is OK, the concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.
No	If the current level LED is not illuminated and the sensor output LEDs do not illuminate or if the current level LED is illuminated red, INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor — Front or Wheel Speed Sensor — Rear in this section. CLEAR the DTCs. REPEAT the self-test. If the current level LED is not illuminated and the sensor output LEDs illuminate green but do not flash, INSPECT the wheel speed sensor tone ring and INSTALL new if necessary. If the tone ring is OK, INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor — Front or Wheel Speed Sensor — Rear in this section. CLEAR the DTCs. REPEAT the self-test.

E5 CHECK THE ABS MODULE OUTPUT

NOTE: Make sure all test connections are made at the wheel speed sensor before the ignition is turned to the ON position.

NOTE: Use a meter with a min/max feature and/or recording capabilities to obtain accurate measurements. The ABS module disables the voltage output immediately if the wheel speed sensor is not present.

- Ignition OFF.
- Connect the meter to the suspect wheel speed sensor electrical connector pins, harness side.
- Enable the min/max or recording feature.
- Ignition ON.
- Check the maximum recorded voltage on the meter.

Is the maximum recorded voltage greater than 10 volts?

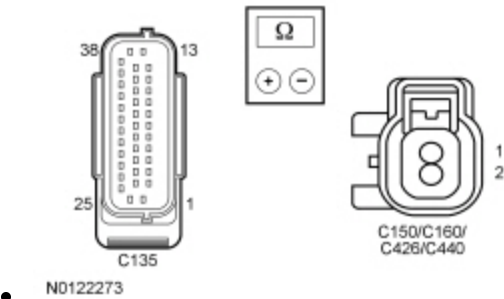
Yes	GO to E7 .
No	GO to E6 .

E6 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR AN OPEN

NOTE: Check both circuits for each DTC.

- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Disconnect: Suspect Wheel Speed Sensor .
- Measure the resistance between the ABS module connector, harness side and the suspect wheel speed sensor connector, harness side as follows:

ABS Module	Circuit	Wheel Speed Sensor
C135 Pin 21	VCA03 (VT/WH)	C150 Pin 1 (LF)
C135 Pin 34	RCA17 (YE)	C150 Pin 2 (LF)
C135 Pin 18	VCA05 (GY/VT)	C160 Pin 1 (RF)
C135 Pin 6	RCA19 (VT)	C160 Pin 2 (RF)
C135 Pin 19	VCA06 (WH/OG)	C426 Pin 1 (RR)
C135 Pin 31	RCA20 (BN)	C426 Pin 2 (RR)
C135 Pin 33	VCA04 (BU/OG)	C440 Pin 1 (LR)
C135 Pin 20	RCA18 (BN/GN)	C440 Pin 2 (LR)



Are the resistances less than 5 ohms?

Yes	GO to E11 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.

E7 CHECK THE WHEEL SPEED SENSOR AND MOUNTING

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
 - **NOTE:** Use a good light source and carefully examine the wheel speed sensor wire. Failure to verify damage in the wheel speed sensor wire can lead to unnecessary installation of a new component.
- Inspect the wheel speed sensor for looseness.

Is the wheel speed sensor and mounting OK?

Yes	GO to E9 .
No	TIGHTEN the wheel speed sensor to specification. REFER to Specifications in this section. CLEAR the DTCs. GO to E8 .

E8 RECHECK THE WHEEL SPEED SENSOR PIDS

- While driving the vehicle at a constant speed, monitor the following PIDs:
 - Left Front Wheel Speed (LF_WSPD)
 - Right Front Wheel Speed (RF_WSPD)
 - Left Rear Wheel Speed (LR_WSPD)
 - Right Rear Wheel Speed (RR_WSPD)

Are all the wheel speed sensor PID values similar?

Yes	The vehicle is OK. The concern may have been caused by a loose wheel speed sensor.
No	GO to E9 .

E9 CHECK THE WHEEL SPEED SENSOR(S) FOR DAMAGE

- Ignition OFF.
- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Remove the wheel speed sensor. Refer to Wheel Speed Sensor — Front or Wheel Speed Sensor — Rear in this section.
- **NOTE:** *Use a good light source and carefully examine the wheel speed sensor wire. Failure to verify damage in the wheel speed sensor wire can lead to unnecessary installation of a new component.*

Inspect the wheel speed sensor and attached wire harness for general damage.

Is the wheel speed sensor and harness OK?

Yes	For DTC C1296 or C1297, INSTALL a new wheel hub and bearing (wheel speed sensor tone rings are part of the wheel hub and bearing assembly). REFER to Section 204-01A or Section 204-01B. For DTC C1298 or C1299, INSTALL a new wheel speed sensor tone ring. REFER to Wheel Speed Sensor Ring — Rear in this section. CLEAR the DTCs. REPEAT the self-test.
No	INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor — Front or Wheel Speed Sensor — Rear in this section. CLEAR the DTC. REPEAT the self-test.

E10 CHECK THE ABS MODULE CONFIGURATION

- Verify that the tires are the correct size for the vehicle and inflated to the specified pressures as indicated on the Vehicle Certification (VC) label.
- Carry out the ABS module configuration. Verify that the vehicle data matches the vehicle options. Refer to Programmable Module Installation (PMI) in Section 418-01.

- Clear the DTCs. Carry out the ABS module on-demand self-test.

Is DTC C1222 still present?

Yes	GO to E11 .
No	The system is operating correctly at this time. The concern may have been caused by an incorrectly configured ABS module.

E11 CHECK FOR CORRECT ABS MODULE OPERATION

- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Check ABS module connector for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: ABS Module [C135](#).
- Make sure the connector seats correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSPECT the wheel speed sensor tone rings for damage. INSTALL new components as necessary. If the tone rings are OK, INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test F: DTC C1266

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The operating voltage required for ABS module isolation valve operation is between 10 and 17 volts.

Fused battery voltage is supplied to the ABS module for the valves by the Battery Junction Box (BJB) .

- DTC C1266 (ABS Valve Power Relay Circuit Failure) — this DTC sets in the ignition cycle if the ABS module detects an open or a short to ground on the valve power feed circuit. DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected.

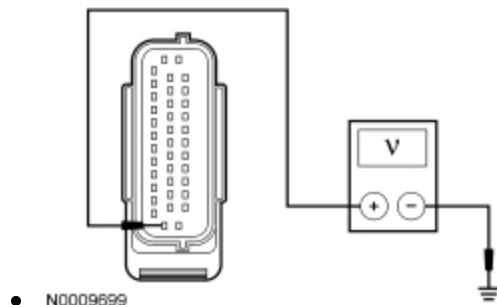
This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- ABS module

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

F1 CHECK FOR VOLTAGE TO THE ABS MODULE

- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Measure the voltage between ABS module [C135](#) Pin 25, circuit SBB36 (GN/RD), harness side and ground.

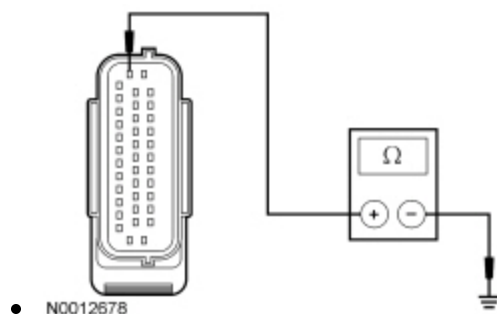


Is the voltage greater than 10 volts?

Yes	GO to F2 .
No	VERIFY that Battery Junction Box (BJB) fuse 36 (30A) is OK. If OK, REPAIR circuit SBB36 (GN/RD). If not OK, REFER to the Wiring Diagrams manual to identify possible causes of the circuit short. CLEAR the DTCs. REPEAT the self-test.

F2 CHECK THE ABS MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between ABS module [C135](#) Pin 38, circuit GD121 (BK/YE), harness side and ground.



Is the resistance less than 0.5 ohm?

Yes	GO to F3 .
No	REPAIR circuit GD121 (BK/YE). CLEAR the DTCs. REPEAT the self-test.

F3 CHECK FOR CORRECT ABS MODULE OPERATION

- Disconnect: ABS Module [C135](#).

- Check ABS module connector for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: ABS Module [C135](#).

Is the concern still present?

Yes	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test G: DTCs C116A, C1440

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The pressure transducer is integral to the Hydraulic Control Unit (HCU) . The pressure transducer sends a signal to the ABS module that varies in relation to the amount of pressure applied to the brake pedal. The more pressure is applied, the greater the signal sent to the ABS module. If a new pressure transducer is needed, then a new HCU must be installed.

When the brake pedal is pressed, the stoplamp switch closes and voltage is sent to the PCM. The PCM then sends a message over the High Speed Controller Area Network (HS-CAN) to the ABS module indicating the brake pedal has been pressed. DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected.

- DTC C116A (ABS Pressure Transducer/Brake Switch Mismatch) — this DTC sets in every ignition cycle if, during normal operation, the stoplamp switch message does not match the HCU transducer, both signals have to agree about whether the brakes are being applied or not.
- DTC C1440 (Pressure Transducer Main/Primary Signal Faulted) — this DTC sets in every ignition cycle if the signal the pressure transducer sends to the ABS module does not agree with the stoplamp switch position.

This pinpoint test is intended to diagnose the following:

- Network communication concern
- Stoplamp switch
- HCU

PINPOINT TEST G : DTCS C116A, C1440

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

G1 INSPECT THE STOPLAMP SWITCH AND VERIFY INSTALLATION

- Inspect the stoplamp switch for obvious signs of damage.
- Remove and install the stoplamp switch. Refer to Stoplamp Switch Removal and Installation in Section 417-01.
- Press the brake pedal and verify the stoplamps operate correctly.

Do the stoplamps operate correctly and is the switch free from any damage?

Yes	GO to G2 .
No	If the stoplamp switch is damaged, INSTALL a new stoplamp switch. REFER to Stoplamp Switch Removal and Installation in Section 417-01. DIAGNOSE the stoplamps. REFER to Stoplamp Switch Diagnosis and Testing in Section 417-01.

G2 MONITOR THE ABS MODULE BRAKE ON/OFF (BOO_ABS) PID

- Connect the scan tool.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- Press and release the brake pedal while monitoring the BOO_ABS PID.

Does the PID display ON with the brake pedal applied and OFF with the brake pedal released?

Yes	GO to G4 .
No	GO to G3 .

G3 MONITOR THE PCM BRAKE ON/OFF (BOO1) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — PCM .
- Press and release the brake pedal while monitoring the BOO1 PID.

Does the PID display ON with the brake pedal applied and OFF with the brake pedal released?

Yes	REFER to Section 418-00 to continue diagnosis of the HS-CAN .
No	REFER to Section 417-01 to continue diagnosis of the stoplamp switch fault.

G4 MONITOR THE INITIAL (BRAKE PEDAL NOT APPLIED) BRAKE PRESSURE USING THE ABS MODULE BRAKE PRESSURE (BRAKPRES) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- With the brake pedal **not** applied, record the pressure displayed by the BRAKPRES PID.

Is the initial brake pressure less than 275 kPa (40 psi) with the brake pedal not applied?

Yes	GO to G5 .
No	INSTALL a new HCU . REFER to Hydraulic Control Unit (HCU) in this section.

G5 MONITOR THE BRAKPRES PID WHILE APPLYING THE BRAKE PEDAL

- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .

- Firmly press and release the brake pedal while monitoring the BRAKPRES PID.

Does the PID increase with firm pressure on the brake pedal and then decrease with the brake pedal released?

Yes	The system is operating correctly at this time. The concern may have been caused by an incorrectly installed stoplamp switch.
No	INSTALL a new HCU . REFER to Hydraulic Control Unit (HCU) in this section.

Pinpoint Test I: DTCs C109D, C109E and C109F

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module uses the vacuum sensor to determine if hydraulic pump intervention is necessary due to a severe loss of vacuum in the booster. The ABS module sends a 5-volt reference voltage to the sensor and the sensor compares the pressure in the booster to the ambient air pressure in the engine compartment and sends a varying voltage back to the ABS module based on the difference between the pressures. Normally, the pressure in the booster is much lower than ambient air pressure. As the pressure in the booster gets higher (closer to matching ambient air pressure), the lower the voltage sent back to the ABS module becomes.

Vehicles equipped with a 3.5L engine are equipped with a camshaft-driven vacuum pump which supplies vacuum to the brake booster. For all other engine types, the vacuum is supplied to the brake booster directly from the engine.

- DTC C109D (Low Vacuum Condition Detected) — this DTC sets in continuous memory and every ignition cycle when the vacuum in the brake booster goes below a predetermined threshold and does not necessarily indicate a concern with the vacuum sensor, the related wiring or the ABS module. This DTC indicates a vacuum concern in the brake booster, engine and any related vacuum hoses/tubes. High altitude locations with low ambient air temperatures can cause a low vacuum condition to exist in the booster until the engine compartment warms up.
- DTC C109E (Vacuum Sensor Circuit Fault) — this DTC sets in continuous memory and every ignition cycle if during normal operation or the ABS module on-demand self-test, the ABS module detects an open circuit, a short to voltage or a short to ground on the vacuum sensor circuits or a failure of the vacuum sensor.
- DTC C109F (Vacuum Sensor Signal Fault) — this DTC sets in continuous memory and every ignition cycle if the vacuum sensor ambient air pressure inlet ports, or the vacuum sensor ports become restricted or clogged.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Vacuum sensor
- Vacuum hoses
- Front hub lock ~~IWE~~ vacuum hose connection
- Brake master cylinder seal
- Check valve grommet
- Vacuum sensor grommet
- Check valve
- Brake vacuum pump
- ABS module

PINPOINT TEST I : DTCS C109D, C109E AND C109F

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

I1 CHECK THE ABS MODULE DTCS

- Review the DTCs recorded during Inspection and Verification.

Is DTC C109D the only recorded ABS module DTC?

Yes	GO to I2 .
No	If DTC C109E or C109F is also present, GO to I6 . For all other DTCs, REFER to the ABS Module DTC Chart in this section.

I2 CHECK DTC RETURNABILITY

- Using the scan tool, clear the ABS module DTCs.
- Ignition OFF.
- Start the engine.
- **NOTE:** *It may take up to 3 minutes for DTC C109D to set.*

Allow the vehicle to idle for a minimum of 4 minutes. Apply and release the brake pedal for a minimum of 3 times per minute with normal pedal force.

- Ignition OFF.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .

Is DTC C109D the only vacuum sensor DTC present?

Yes	GO to I3 .
No	If DTC C109E or C109F is also present, GO to I6 . For all other DTCs, REFER to the ABS Module DTC Chart in this section. If no DTC is set, the system is operating correctly as this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTC. REPEAT the self-test.

I3 CHECK THE BRAKE BOOSTER AND VACUUM COMPONENT

- Ignition OFF.
- Inspect the following components for signs of obvious damage:
 - Check valve grommet
 - Vacuum sensor grommet
 - Brake master cylinder seal
 - Check valve
 - Vacuum hoses
 - Brake booster
 - Front hub lock ~~IWE~~ vacuum hose connection

Are the components listed free from any damage?

Yes	GO to 14 .
No	INSTALL new components as necessary, then GO to 14 .

I4 CHECK THE BRAKE BOOSTER

- Carry out the Brake Booster Component Test. Refer to Section 206-00.

Was the component test successful?

Yes	GO to 15 .
No	INSTALL new components as directed by the component test.

I5 VERIFY ABS MODULE BRAKE BOOSTER VACUUM LEVEL (BBVACLVL) PID

NOTE: *Running the engine with the vacuum sensor removed from the brake booster sets Diagnostic Trouble Codes (DTCs) in the ABS module. Clear all Diagnostic Trouble Codes (DTCs) once the repair is complete.*

- Start the engine and allow the to idle for at least 20 seconds.
- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- Observe and record the ABS module BBVACLVL PID display.
- Ignition OFF.
- Remove the vacuum sensor from the brake booster (leave the electrical connector connected).
- Connect a vacuum gauge to the vacuum sensor port.
- Start the engine and allow the to idle for at least 20 seconds.
- Observe and record the vacuum gauge display.
- Ignition OFF.
- Disconnect the vacuum gauge from the brake booster and install the vacuum sensor.

Does the PID display match the recorded vacuum display ± 10 kPa (3 in-Hg)?

Yes	The condition causing the concern is not present at this time. The concern may be temperature related. ALLOW the vehicle to remain outside overnight and RETEST the vacuum system.
No	INSTALL a new brake booster vacuum sensor. REFER to Section 206-07. CLEAR the DTCs. TEST the system for normal operation.

I6 CHECK THE BRAKE BOOSTER VACUUM SENSOR

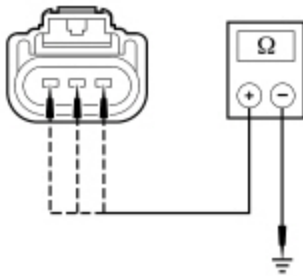
- Inspect the vacuum sensor ambient air pressure inlet ports.

Are both ports free of contamination, debris and foreign material?

Yes	GO to 17 .
No	REPAIR or INSTALL a new vacuum sensor as necessary. REFER to Section 206-07. CLEAR the DTCs. REPEAT the self-test.

I7 CHECK THE BRAKE BOOSTER VACUUM SENSOR CIRCUITS FOR A SHORT TO GROUND

- Disconnect: ABS Module [C135](#).
- Disconnect: Brake Booster Vacuum Sensor [C149](#).
- Measure the resistance between ground and:
 - brake booster vacuum sensor [C149](#) Pin 1, circuit VCA22 (GN/BN), harness side.
 - brake booster vacuum sensor [C149](#) Pin 2, circuit GD120 (BK/GN), harness side.
 - brake booster vacuum sensor [C149](#) Pin 3, circuit CCA22 (GY/YE), harness side.



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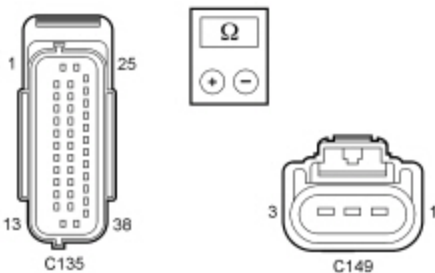
Is the resistance greater than 10,000 ohms?

Yes	GO to 18 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.

I8 CHECK THE BRAKE BOOSTER VACUUM SENSOR CIRCUITS FOR AN OPEN

- Measure the resistance between the ABS module [C135](#), harness side and the brake booster vacuum sensor [C149](#), harness side as follows:

ABS Module	Circuit	Vacuum Sensor
C135 Pin 10	VCA22 (GN/BN)	C149 Pin 1
C135 Pin 28	GD120 (BK/GN)	C149 Pin 2
C135 Pin 2	CCA22 (GY/YE)	C149 Pin 3



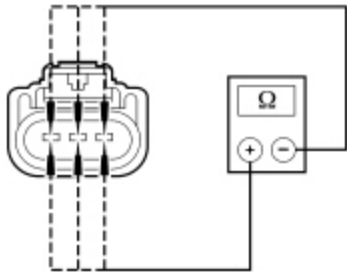
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Are the resistances less than 5 ohms?

Yes	GO to I9 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.

I9 CHECK THE BRAKE BOOSTER VACUUM SENSOR CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between:
 - brake booster vacuum sensor [C149](#) Pin 1, circuit VCA22 (GN/BN), harness side and brake booster vacuum sensor [C149](#) Pin 2, circuit GD120 (BK/GN), harness side.
 - brake booster vacuum sensor [C149](#) Pin 1, circuit VCA22 (GN/BN), harness side and brake booster vacuum sensor [C149](#) Pin 3, circuit CCA22 (GY/YE), harness side.
 - brake booster vacuum sensor [C149](#) Pin 2, circuit GD120 (BK/GN), harness side, and brake booster vacuum sensor [C149](#) Pin 3, circuit CCA22 (GY/YE), harness side.



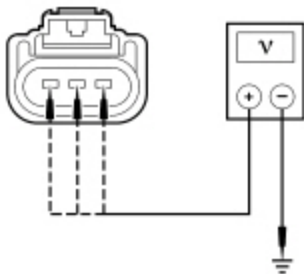
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Are the resistances greater than 10,000 ohms?

Yes	GO to I10 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.

I10 CHECK THE ACTIVE BRAKE BOOSTER VACUUM SENSOR CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition ON.
- Measure the voltage between ground and:
 - brake booster vacuum sensor [C149](#) Pin 1, circuit VCA22 (GN/BN), harness side.
 - brake booster vacuum sensor [C149](#) Pin 2, circuit GD120 (BK/GN), harness side.
 - brake booster vacuum sensor [C149](#) Pin 3, circuit CCA22 (GY/YE), harness side.



N0101667

Is any voltage present?

Yes	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.
No	GO to I11 .

I11 CHECK THE BRAKE BOOSTER VACUUM SENSOR CONNECTOR

- Ignition OFF.
- Check brake booster vacuum sensor connector for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: Brake Booster Vacuum Sensor [C149](#).
- Make sure the connector seats correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new vacuum sensor. REFER to Section 206-07. CLEAR the DTCs. REPEAT the self-test. If DTC C109E or C109F return, GO to I12 .
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

I12 CHECK THE ABS MODULE CONNECTORS

- Ignition OFF.
- Disconnect: ABS Module [C135](#).
- Check ABS module connector for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: ABS Module [C135](#).
- Make sure the connector seats correctly, then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

Hill Descent Control™ can be activated by momentarily pressing the Hill Descent Control™ switch at speeds from 3 km/h (2 mph) and 16 km/h (10 mph), but the system stays in control up to 32 km/h (20 mph). At speeds between 32 km/h (20 mph) and 64 km/h (40 mph), Hill Descent Control™ does not set or maintain speeds. Hill Descent Control™ automatically disables itself above 64 km/h (40 mph).

When the Hill Descent Control™ switch is pressed, a circuit inside of the switch is closed, ground is provided to the Instrument Panel Cluster (IPC) disabling or enabling the Hill Descent Control™ .

DTC C2809 can be set by manually pressing and holding the Hill Descent Control™ switch for more than 30 seconds.

- DTC C2809 (Hill Descent Control Switch Stuck) — this DTC sets if the ABS module receives the HILL_DESC_SW message from the High Speed Controller Area Network (HS-CAN) for greater than 30 seconds.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Hill Descent Control™ switch
- ABS module

PINPOINT TEST J : THE HILL DESCENT CONTROL SYSTEM IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

J1 CHECK FOR DTCS

- Connect the scan tool.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — All CMDTCs .

Are any ABS module, IPC or communication DTCs present?

Yes	For ABS module DTC C2809, GO to J2 . For all other ABS module DTCs, REFER to the ABS Module DTC Chart. For IPC DTCs, REFER to the Section 413-01.
No	System is operating correctly at this time.

J2 VISUALLY INSPECT THE HILL DESCENT CONTROL™ SWITCH

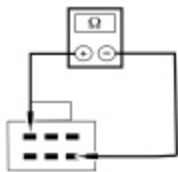
- Inspect the Hill Descent Control™ switch for damage.
- Verify that the Hill Descent Control™ switch does not stick.

Is the Hill Descent Control™ switch damaged or sticking?

Yes	INSTALL a new Hill Descent Control™ switch by removing the auxiliary switch assembly. REFER to the exploded view in Section 501-12. PRESS the tab on each side of the switch to remove it from the bezel. After replacing the switch, CLEAR the DTCs and REPEAT the self-test. TEST the system for normal operation.
No	GO to J3 .

J3 ISOLATE THE HILL DESCENT CONTROL™ SWITCH

- Ignition OFF.
- Disconnect: Hill Descent Control™ Switch [C3394](#).
- Measure the resistance between Hill Descent Control™ switch pins 1 and 6, component side, while pressing and releasing the switch.



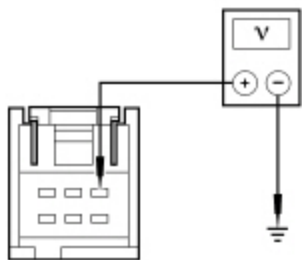
• N0122269

Is the resistance less than 5 ohms with the switch pressed and greater than 10,000 ohms with the switch released?

Yes	If DTC C2809 was recorded, GO to J4 . For all other conditions, GO to J5 .
No	INSTALL a new Hill Descent Control™ switch by removing the auxiliary switch assembly. REFER to the exploded view in Section 501-12. PRESS the tab on each side of the switch to remove it from the bezel. After replacing the switch, CLEAR the DTCs and REPEAT the self-test. TEST the system for normal operation.

J4 CHECK THE HILL DESCENT CONTROL™ SWITCH SIGNAL CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: IPC [C220](#) .
- Ignition ON.
- Measure the voltage between Hill Descent Control™ switch [C3394](#) Pin 1, circuit CET59 (GY/VT), harness side and ground.



• N0101067

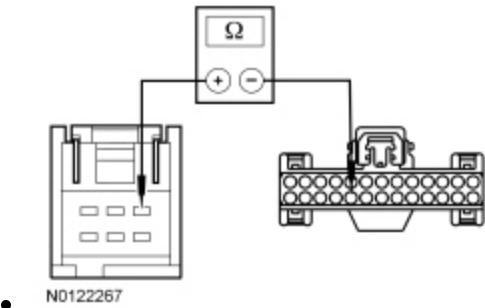
Is any voltage present?

Yes	REPAIR circuit CET59 (GY/VT). CLEAR the DTCs. REPEAT the self-test.
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No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST for normal operation.
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J5 CHECK THE HILL DESCENT CONTROL™ SWITCH SIGNAL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: IPC [C220](#) .
- Measure the resistance between Hill Descent Control™ switch [C3394](#) Pin 1, circuit CET59 (GY/VT), harness side and IPC [C220](#) Pin 22, circuit CET59 (GY/VT), harness side.

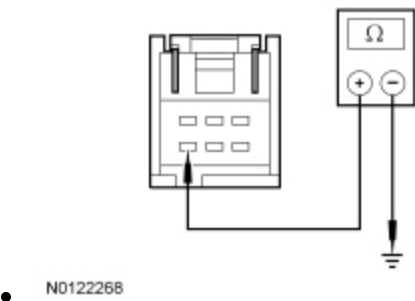


Is the resistance less than 5 ohms?

Yes	GO to J6 .
No	REPAIR circuit CET59 (GY/VT). TEST the system for normal operation.

J6 CHECK THE HILL DESCENT CONTROL™ SWITCH GROUND CIRCUIT

- Ignition ON.
- Measure the resistance between Hill Descent Control™ switch [C3394](#) Pin 6, circuit GD138 (BK/WH), harness side and ground.



Is the resistance less than 5 ohms?

Yes	GO to J7 .
No	REPAIR the open in circuit GD138 (BK/WH). CLEAR the DTCs. REPEAT the self-test.

J7 CHECK FOR CORRECT IPC OPERATION

- Check IPC connector:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the IPC connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC . REFER to Section 413-01. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test K: DTC U0100

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

- DTC U0100 (Lost Communication With ECM/PCM) — this DTC sets in continuous memory if the brake pedal position switch, and traction control messages that should be received from the ECM/PCM are missing for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- PCM
- ABS module

PINPOINT TEST K : DTC U0100

K1 VERIFY THE SCAN TOOL COMMUNICATES WITH THE PCM

- Connect the scan tool.
- Check that a vehicle session can be established using the scan tool.

Can a vehicle session be established?

Yes	GO to K2 .
No	REFER to Section 418-00 to diagnose no communication with the PCM.

K2 CHECK THE ABS MODULE CONTINUOUS MEMORY DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .
- Clear the DTCs.
- Wait 10 seconds.

- Repeat the ABS module self-test.

Is DTC U0100 retrieved again?

Yes	GO to K3 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

K3 RETRIEVE THE RECORDED DTCS FROM THE PCM KOEO SELF-TEST

- Check for recorded DTCS from the PCM Key ON Engine OFF (KOEO) self-test.

Is DTC P0562 or P0563 recorded?

Yes	For PCM DTCS, REFER to Section 303-14.
No	GO to K4 .

K4 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE SELF-TEST

- Check for recorded DTCS from the ABS module self-test.

Is DTC B1676 recorded?

Yes	GO to Pinpoint Test A .
No	GO to K5 .

K5 CHECK FOR DTC U0100 SET IN OTHER MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — All CMDTCs .
- Retrieve the continuous memory DTCS from all modules.

Is DTC U0100:00 or U0100 set in more than 1 module?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.

Pinpoint Test L: DTC U0114

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

- DTC U0114 (Lost Communication with Transfer Case Control Module (TCCM)) — this DTC sets in continuous memory if the Four-Wheel Drive (4WD) message that should be received from the Transfer Case Control Module (TCCM) is missing for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- ~~TCCM~~
- ABS module

PINPOINT TEST L : DTC U0114

L1 VERIFY THE CUSTOMER CONCERN	
• Ignition ON. • Verify there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to L2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
L2 CHECK THE COMMUNICATION NETWORK	
• Using a scan tool, carry out a Network Test.	
Does the TCCM pass the network test?	
Yes	GO to L3 .
No	REFER to Section 418-00.
L3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE AND TCCM SELF-TESTS	
• Enter the following diagnostic mode on the scan tool: Self Test — ABS Module . • Enter the following diagnostic mode on the scan tool: Self Test — TCCM .	
Is DTC B1676 (or P0562:00) or DTC P0563:00 recorded?	
Yes	For ABS module DTC B1676, GO to Pinpoint Test A . For TCCM DTCs, REFER to Section 308-07A.
No	GO to L4 .
L4 RECHECK THE ABS MODULE DTCS	
• Clear the DTCs. • Ignition OFF.	

- Wait 10 seconds.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .

Is DTC U0114 still present?

Yes	GO to L5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

L5 CHECK FOR DTC U0114 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self Test — All ~~CMD~~DTCs .

Is DTC U0114 set in more than 1 module?

Yes	INSTALL a new TCCM TCCM . REFER to Section 308-07A. CLEAR the DTCs. REPEAT the ABS module self-test.
No	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.

Pinpoint Test M: DTC U0131

Normal Operation

- DTC U0131 (Lost Communication With Power Steering Control Module (VAPS) — this DTC sets in continuous memory and on-demand if the steering sensor message that should be received from the Power Steering Control Module (PSCM) is missing for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- ~~PSCM~~ PSCM
- ABS module

PINPOINT TEST M : DTC U0131

M1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to M2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

M2 CHECK THE COMMUNICATION NETWORK

- Using a scan tool, carry out a Network Test.

Does the PSCM pass the network test?

Yes	GO to M3 .
No	REFER to Section 418-00.

M3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE AND PSCM SELF-TESTS

- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .
- Enter the following diagnostic mode on the scan tool: Self Test — PSCM .

Is DTC B1676 (or U3003:16) or DTC U3003:17 recorded?

Yes	For ABS module DTC B1676, GO to Pinpoint Test A . For PSCM DTCs, REFER to Section 211-02.
No	GO to M4 .

M4 RECHECK THE ABS MODULE DTCS

- Clear the DTCs.
- Ignition OFF.
- Wait 10 seconds.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .

Is DTC U0131 still present?

Yes	GO to M5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

M5 CHECK FOR DTC U0131 SET IN OTHER MODULES

- Clear the DTCs.

- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self Test — All CMDTCs .

Is DTC U0131 set in more than 1 module?

Yes	REFER to the Steering System and NVH Symptom Chart, A Module Cannot Communicate With The PSCM . INSTALL a new EPAS. Refer to the appropriate Removal and Installation procedure in Section 211-02.
No	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.

Pinpoint Test N: DTC U0131

Normal Operation

- **NOTE:** *Vehicles with a 6.2L engine are equipped with a Steering Column Control Module (SCCM) .*

DTC U0131 (Lost Communication With Power Steering Control Module (VAPS) — this DTC sets in continuous memory and on-demand if the steering sensor message that should be received from the SCCM is missing for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- SCCM .
- ABS module

PINPOINT TEST N : DTC U0131

N1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify there is an observable symptom present.

Is an observable symptom present?

Yes	GO to N2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

N2 CHECK THE COMMUNICATION NETWORK

- Using a scan tool, carry out a Network Test.

Does the SCCM pass the network test?

Yes	GO to N3 .
No	REFER to Section 418-00.

N3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE AND SCCM SELF-TESTS

- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .
- Enter the following diagnostic mode on the scan tool: Self Test — ~~SCCM~~ .

Is DTC B1676 (or B11D9:16) or DTC B11D9:17 recorded?

Yes	For ABS module DTC B1676, GO to Pinpoint Test A . For SCCM DTCs, REFER to Section 211-05.
No	GO to N4 .

N4 RECHECK THE ABS MODULE DTCS

- Clear the DTCs.
- Ignition OFF.
- Wait 10 seconds.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .

Is DTC U0131 still present?

Yes	GO to N5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

N5 CHECK FOR DTC U0131 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self Test — All ~~CMDTCs~~ .

Is DTC U0131 set in more than 1 module?

Yes	INSTALL a new SCCM . REFER to Section 211-05. CLEAR the DTCs. REPEAT the ABS module self-test.
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No	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.
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Pinpoint Test O: DTCs U0140

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

- DTC U0140 (Lost Communication With Body Control Module (GEM)) — this DTC sets in the ABS module if it cannot transmit to or receive messages from the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- ABS module
- BCM

PINPOINT TEST O : DTC U0140

O1 VERIFY THE CUSTOMER CONCERN

- Ignition ON.
- Verify that there is an observable symptom present.

Is an observable symptom present?

Yes	GO to O2 .
No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

O2 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- Using a scan tool, carry out a Network Test.

Does the BCM pass the network test?

Yes	GO to O3 .
No	REFER to Section 418-00.

O3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE AND BCM SELF-TESTS

- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .
- Enter the following diagnostic mode on the scan tool: Self Test — BCM .

Is DTC B1676 (or U3003:16) or DTC U3003:17 recorded?

Yes	For ABS module DTC B1676, GO to Pinpoint Test A . For BCM DTCs, REFER to Section 419-10.
No	GO to O4 .

O4 CHECK FOR DTC U0140 OR U0140:00 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self Test — All CMDTCs .

Is DTC U0140 or U0140:00 set in more than 1 module?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation. CLEAR the DTCs. REPEAT the ABS module self-test
No	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.

Pinpoint Test P: DTCs C1280, C1282, C1516, C1963, C1975, C2770

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module uses information received from several sensors (wheel speed sensors, steering wheel rotation sensor, stability/traction control system sensors, etc.) to determine if vehicle stability, traction or roll warrants ESC or RSC® interaction. If an event is going to occur, the ABS module adjusts the brake torque at specific wheels and the PCM reduces engine torque to make the vehicle more stable. Once the ABS module has determined that the stability event has been corrected, it returns the brake and engine torque to normal and continues to monitor the vehicle. DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected. The Restraints Control Module (RCM) measures the yaw rate and vehicle acceleration, then sends the signal through a secondary, dedicated Controller Area Network (CAN) to the ABS module. The ABS module uses this information for AdvanceTrac® and RSC® system operation. The ABS module also compares this information to information it receives from other sensors (brake pedal input, steering wheel) to determine how valid the information is and to determine if an instability event is about to occur.

The RCM and the ABS module communicate along circuits VCA23 (BU/WH) and VCA24 (GN/OG). An open circuit or a short to power or ground on any one of these circuits will cause the ABS module to set a DTC. Also, if the RCM is not mounted in the correct position, with the electrical connector facing the driver-side of the vehicle, the ABS module will also set DTCs.

DTC Description	Fault Trigger Condition
• C1280 — Yaw Rate Sensor Signal Fault • C1282 — Lateral Accelerometer Signal Fault • C1516 — Roll Rate Sensor Signal Fault • C2770 — Longitudinal Acceleration Sensor Signal Fault	If the ABS module receives a message from the <u>RCM</u> that is out of the normal operating range or one that does not agree with other sensor information, the appropriate signal fault DTC will be set.
• C1963 — Stability Control Inhibit Warning • C1975 — IVD Plausibility Failure	If conditions or DTCs exist that prevent the AdvanceTrac® and <u>RSC</u> ® systems from operating, DTCs C1963 and/or C1975 will be set. This DTC is usually set due to the steering wheel rotation sensor or the <u>RCM</u> being damaged or installed incorrectly. If DTCs are present in the <u>RCM</u> or <u>PSCM</u> , diagnose those first.

^a

^aInteractive Vehicle Dynamics

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Wheels or tires
- RCM

PINPOINT TEST P : DTCS C1280, C1282, C1516, C1963, C1975, C2770

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may cause damage to the connector.
NOTE: <i>Supplemental Restraint System (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the <u>SRS</u> .</i>
NOTE: <i>The <u>SRS</u> must be fully operational and free of faults before releasing the vehicle to the customer.</i>
P1 CHECK FOR RETURNING ABS MODULE DTCS
• Ignition ON. • Using a diagnostic scan tool, clear the ABS module DTCs.

- Ignition OFF.
- Ignition ON.
- Test drive the vehicle above 12 km/h (7 mph) and make at least 1 left turn and 1 right turn.
- Using a diagnostic scan tool, carry out the ABS module self-test.
- Record all ABS module DTCs.

Are any DTCs present in the ABS module?

Yes	If DTC C1280, C1282, C1516, C1963, C1975 or C2770 is present, GO to P2 . For all other ABS module DTCs, GO to the ABS Module DTC Chart.
No	The fault is not present and cannot be recreated at this time. CHECK <u>QASIS</u> for any applicable <u>TSBs</u> . If a <u>TSB</u> exists for this concern, DISCONTINUE this test and FOLLOW the <u>TSB</u> instructions. If no <u>TSBs</u> address this concern, INSPECT all in-line connectors and the wire harness between the <u>RCM</u> and the ABS module for corrosion, loose terminals, spread terminals, loose wire connections, frayed wire connections, pinched wire harnesses and cut or pierced wire harnesses. REPAIR as necessary.

P2 CHECK THE WHEELS AND TIRES

- Inspect the wheels and tires for any damage or excessive wear.
- Verify all wheels and tires are the same size and match the size indicated on the VC label.
- Verify the inflation pressure of all 4 tires is correct as indicated on the VC label.

Are the wheels and tires OK?

Yes	GO to P3 .
No	INSTALL the correct size wheel or tire as necessary. ADJUST tire pressure as necessary.

P3 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- **NOTE:** The Network Test does not test the dedicated CAN communications between the ABS module and RCM.

Using the diagnostic scan tool, carry out the Network Test.

Does the RCM pass the Network Test?

Yes	GO to P4 .
No	DIAGNOSE the RCM does not communicate with the diagnostic scan tool. REFER to Section 418-00.

P4 CHECK THE ABS MODULE CMDTCS

- Using a diagnostic scan tool, carry out the ABS module self-test.
- Retrieve and record all ABS module DTCs.
- Clear the ABS module DTCs.
- Ignition OFF.
- Ignition ON.
- Using a diagnostic scan tool, carry out the ABS module self-test.

Are any DTCs present in the ABS module?

Yes	If DTC C1280, C1282, C1516, C1963, C1975 or C2770 is present, GO to P5 . For all other ABS module DTCs, GO to the ABS Module DTC Chart.
No	GO to P6 .

P5 CHECK FOR RCM DTCS

- Using a diagnostic scan tool, carry out the **RCM** self-test.

Are any DTCs present in the **RCM** ?

Yes	DIAGNOSE all RCM DTCs. REFER to Section 501-20B.
No	For vehicles equipped with a 6.2L engine, GO to P7 . For all other vehicles, GO to P6 .

P6 CHECK FOR PSCM DTCS

- Using a diagnostic scan tool, carry out the **PSCM** self-test.

Are any DTCs present in the **PSCM** ?

Yes	DIAGNOSE all PSCM DTCs. REFER to Section 211-00A.
No	GO to P7 .

P7 VERIFY STABILITY CONTROL SENSOR CALIBRATION

- Using a diagnostic scan tool, carry out the **RCM** self-test.
- Ignition OFF.
- Ignition ON.
- Test drive the vehicle above 12 km/h (7 mph) and make at least 1 left turn and 1 right turn.

Is the Advance Trac® warning indicator illuminated?

Yes	Using a diagnostic scan tool, CARRY OUT the ABS module self-test. If DTC C1280, C1282, C1516, C1963, C1975 or C2770 is present, GO to P8 . For all other ABS module DTCs, GO to the ABS Module DTC Chart.
No	The condition which caused the concern is no longer present. The concern was most likely due to an incomplete calibration or a failed calibration of the stability control sensors.

P8 CHECK THE ABS MODULE PID VALUES

NOTE: The vehicle must be on level ground and at a complete standstill. Any vehicle movement results in false values for this test.

- Ignition OFF.
- Ignition ON.
- Using a diagnostic scan tool, monitor the following ABS module PIDs :
 - Yaw rate (YAW_RATE_2)
 - Lateral acceleration (LAT_ACCL_2)
 - Longitudinal acceleration (LONG_ACCL)
 - Roll rate (ROLLRAT)

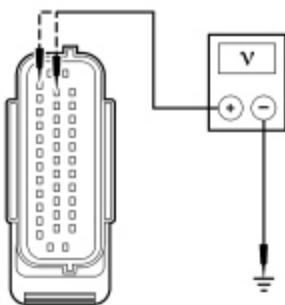
Is the YAW_RATE_2 PID value between -3.5 and 3.5, the LAT_ACCL_2 PID value between -0.125 and 0.125, the LONG_ACCL PID value between -0.15 and 0.15, and the ROLLRAT PID value between -3.5 and 3.5?

Yes	GO to P13 .
No	GO to P9 .

P9 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO VOLTAGE

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.
- Wait one minute then disconnect RCM [C310A](#) and [C310B](#).
- Disconnect ABS module [C135](#).
- Ignition ON.
- Measure the voltage between ground and:
 - ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.
 - ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.

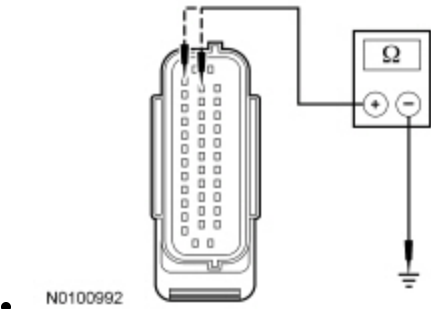


Is any voltage present?

Yes	REPAIR the circuit. CLEAR all C MDTC s . PERFORM a road test. REPEAT the self test.
No	GO to P10 .

P10 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between ground and:
 - ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.
 - ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.

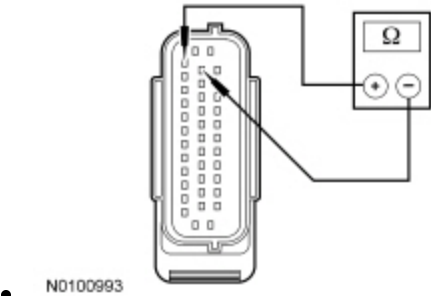


Is the resistance greater than 10,000 ohms?

Yes	GO to P11 .
No	REPAIR the circuit. CLEAR all C MDTC s . PERFORM a road test. REPEAT the self test.

P11 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side and ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.



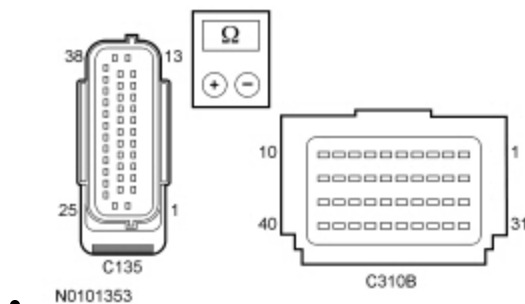
Is the resistance greater than 10,000 ohms?

Yes	GO to P12 .
No	REPAIR the affected circuit(s). CLEAR all C MDTC s . PERFORM a road test. REPEAT the self test.

P12 CHECK THE DEDICATED CAN CIRCUITS FOR AN OPEN

- Measure the resistance between the ABS module [C135](#), harness side and the RCM [C310B](#), harness side as indicated in the following chart:

ABS Module	Circuit	RCM
C135 Pin 37	VCA23 (BU/WH)	C310B Pin 20
C135 Pin 24	VCA24 (GN/OG)	C310B Pin 19



Are the resistances less than 5 ohms?

Yes	GO to P13 .
No	REPAIR the affected circuit(s). CLEAR all CMDTCs . PERFORM a road test. REPEAT the self-test.

P13 CHECK THE RCM INSTALLATION AND MOUNTING SURFACE

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.
- Wait one minute then disconnect RCM [C310A](#) and [C310B](#) (if not previously disconnected).
- Inspect the RCM installation and make sure the fasteners are fully seated and tightened to specifications. Refer to Restraints Control Module (RCM) Removal and Installation in Section 501-20B.
- Inspect the mounting surface for damage, corrosion or foreign material.

Is the RCM installed correctly and is the mounting surface clean and free from damage?

Yes	GO to P14 .
No	CLEAN or REPAIR the mounting surface or correctly REINSTALL the RCM as necessary, REFER to Section 501-20B. CLEAR all CMDTCs . PERFORM a road test. REPEAT the self test.

P14 CHECK THE FOR CORRECT RCM OPERATION

- Ignition OFF.

- Disconnect ABS module [C135](#) and related in-line connectors (if not previously disconnected).
- Inspect the RCM and ABS module electrical connectors and wire harnesses for:
 - corrosion
 - loose or spread terminals
 - loose or frayed wire connections at terminals
 - pushed-out or bent pins
 - pinched, cut or pierced wires
- Repair any wiring or electrical connector concerns found.
- Connect ABS module [C135](#). Make sure it seats and latches correctly.
- Connect RCM [C310A](#) and [C310B](#). Make sure they seat and latch correctly.
- Test drive the vehicle above 12 km/h (7 mph) and make at least 1 left turn and 1 right turn.

Is the Advance Trac® warning indicator illuminated?

Yes	CHECK QASIS for any applicable TSB s . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW the TSB instructions. If no TSB s address this concern, DIAGNOSE the RCM concern. Section 501-20B.
No	The system is operating correctly at this time. The concern may have been caused by a wiring harness or electrical connector issue. ADDRESS the root cause of any wire harness or connector issues.

Pinpoint Test Q: DTC U0155

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

- DTC U0155 (Lost Communication With Instrument Panel Cluster (IC) Control Module) — this DTC sets in continuous memory and on-demand if the traction control switch information, or other sensor messages from the ~~IPC~~ are not received for 5 seconds or longer.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- ~~IPC~~
- ABS module

PINPOINT TEST Q : DTC U0155

Q1 VERIFY THE CUSTOMER CONCERN	
• Ignition ON. • Verify that there is an observable symptom present.	
Is an observable symptom present?	
Yes	GO to Q2 .

No	The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.
-----------	--

Q2 CHECK THE COMMUNICATION NETWORK

- Using a scan tool, carry out a Network Test.

Does the IPC pass the network test?

Yes	GO to Q3 .
No	REFER to Section 418-00.

Q3 RETRIEVE THE RECORDED DTCS FROM THE ABS MODULE AND IPC SELF-TESTS

- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .
- Enter the following diagnostic mode on the scan tool: Self Test — IPC .

Is ABS module DTC B1676 or IPC DTC U3003:16 or DTC U3003:17 received?

Yes	For ABS module DTC B1676, GO to Pinpoint Test A . For <u>IPC</u> DTCs, REFER to Section 413-01.
No	GO to Q4 .

Q4 RECHECK THE ABS MODULE DTCS

- Clear the DTCs.
- Ignition OFF.
- Wait 10 seconds.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .

Is DTC U0155 still present?

Yes	GO to Q5 .
No	The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

Q5 CHECK FOR DTC U0155 OR U0155:00 SET IN OTHER MODULES

- Clear the DTCs.
- Ignition OFF.
- Ignition ON.

- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self Test — All CMDTCs .

Is DTC U0155:00 set in more than 1 module?

Yes	INSTALL a new IPC . REFER to Section 413-01. CLEAR the DTCs. REPEAT the ABS module self-test.
No	INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module in this section. TEST the system for normal operation.

Pinpoint Test R: The Stability/Traction Control System is Inoperative or Cannot be Disabled

Refer to Wiring Diagrams Cell [42](#) , Instrument Cluster for schematic and connector information.

Normal Operation

The stability/traction control switch is a normally open switch. When the switch is pressed, the Instrument Panel Cluster (IPC) senses ground through the switch and sends a message to the ABS module along the High Speed Controller Area Network (HS-CAN) to disable the stability/traction control system. The IPC also illuminates the traction control message center indicator.

NOTE: *If a MyKey® restricted key is in use with the AdvanceTrac® always on feature configured to on, the traction control system cannot be disabled. An admin key must be used in order to enable and disable the traction control system .*

- DTC C2775 (TC/AYC ON/OFF Switch Stuck Failure) — this DTC sets during the key cycle if the IPC senses a short to ground on stability/traction control switch circuit, if the stability/traction control switch is pressed for longer than 30 seconds, or if the stability/traction control switch is pressed during the on-demand self-test.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- MyKey® restricted key in use
- Stability/traction control switch
- IPC

PINPOINT TEST R : THE STABILITY/TRACTION CONTROL SYSTEM IS INOPERATIVE OR CANNOT BE DISABLED

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

R1 VERIFY THAT THE KEY IN USE IS NOT MYKEY® RESTRICTED

- Ignition ON.
- Attempt to disable the Electronic Stability Control (ESC) system by pressing the stability/traction control switch.

Is ADVTRAC ON MyKey® SETTING displayed in the message center?

Yes	INFORM the customer that the key in use is MyKey® restricted. If necessary, REFER to the Owner's Literature for information regarding the programming of MyKey® features.
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No	GO to R2 .
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R2 CHECK FOR ABS MODULE DTCS

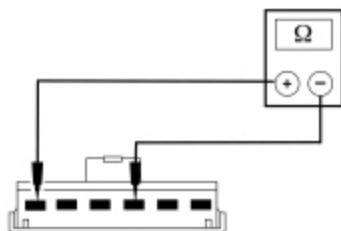
- Attempt to disable the ESC system by pressing the stability/traction control switch.
- Connect the scan tool.
- Enter the following diagnostic mode on the scan tool: Self Test — ABS Module .

Is DTC U0073, U0155 or U0423 present?

Yes	For DTC U0073, REFER to Section 418-00 to diagnose the HS-CAN . For DTC U0155, GO to Pinpoint Test Q . For DTC U0423, CARRY OUT a self-test of the IPC . REFER to Section 413-01 to diagnose the IPC concern.
No	GO to R3 .

R3 ISOLATE THE STABILITY/TRACTION CONTROL SWITCH

- Ignition OFF.
- Disconnect: Stability/Traction Control Switch [C2355](#).
- Measure the resistance between stability/traction control switch pins 1 and 4, component side, while pressing and releasing the switch.



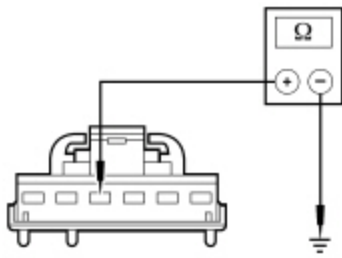
• N0122274

Is the resistance less than 5 ohms with the switch pressed and greater than 10,000 ohms with the switch released?

Yes	GO to R4 .
No	INSTALL a new stability/traction control switch. REFER to Stability/Traction Control Switch in this section. TEST the system for normal operation.

R4 CHECK THE SWITCH INPUT CIRCUIT FOR A SHORT TO GROUND

- Disconnect: Battery Negative Cable .
- Disconnect: IPC [C220](#) .
- Measure the resistance between stability/traction control switch [C2355](#) Pin 4, circuit CCA15 (YE/GY), harness side and ground.



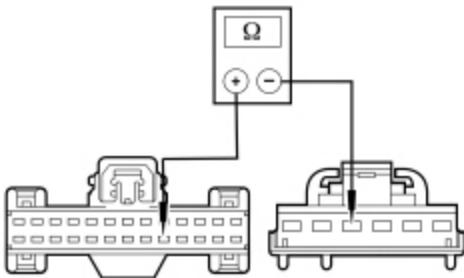
• N0089363

Is the resistance greater than 10,000 ohms?

Yes	GO to R5 .
No	REPAIR circuit CCA15 (YE/GY). TEST the system for normal operation.

R5 CHECK THE SWITCH INPUT CIRCUIT FOR AN OPEN

- Measure the resistance between stability/traction control switch [C2355](#) Pin 4, circuit CCA15 (YE/GY), harness side and ~~IPC~~ [C220](#) Pin 18, circuit CCA15 (YE/GY), harness side.



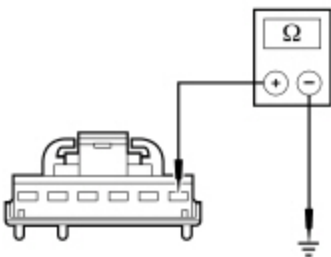
• N0120950

Is the resistance less than 5 ohms?

Yes	GO to R6 .
No	REPAIR circuit CCA15 (YE/GY). TEST the system for normal operation.

R6 CHECK THE SWITCH GROUND CIRCUIT FOR AN OPEN

- Measure the resistance between stability/traction control switch [C2355](#) Pin 1, circuit GD133 (BK), harness side and ground.



• N0105102

Is the resistance less than 5 ohms?

Yes	GO to R7 .
No	REPAIR circuit GD133 (BK). TEST the system for normal operation.

R7 CHECK FOR CORRECT IPC OPERATION

- Check IPC [C220](#) for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the IPC connector and make sure it seats correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new IPC . REFER to Section 413-01. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test S: Poor Vehicle Tracking During Anti-Lock Function

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module uses input from several different sensors (wheel speed sensors, Brake Pedal Position (BPP) switch, sensor cluster, etc.) to determine if an ABS event is necessary. If an ABS event is necessary, the ABS module commands the hydraulic pump motor on and the Hydraulic Control Unit (HCU) to actuate the appropriate solenoids. During an ABS event, the front and rear end of the vehicle should track true to the direction of travel along the vehicle centerline unless driver input dictates otherwise. Other items that could cause uneven braking during an ABS event are; an internal issue with the HCU , an issue with the base brake system (caliper, brake pads, etc.) or worn out suspension components.

The operating voltage required for ABS module, hydraulic pump and isolation valve operation is between 10 and 17 volts.

This pinpoint test is intended to diagnose the following:

- Air in hydraulic brake system
- Low tire pressure
- Base brake system
- HCU

PINPOINT TEST S : POOR VEHICLE TRACKING DURING ANTI-LOCK FUNCTION

S1 VERIFY VEHICLE TRACKING DURING NON-ABS BRAKING

- **NOTE:** Before road testing the vehicle, check the tire pressure and confirm tire pressures are set to the pressure indicated on the [VC](#) label.
- Test drive the vehicle and verify vehicle tracking during normal braking.

Does the vehicle track correctly during normal braking?

Yes	GO to S2 .
No	The condition can be caused by a base brake concern or vehicle alignment. REFER to Section 206-00 to verify the base brake system and REFER to Section 204-00 to verify suspension components.

S2 BLEED THE BRAKE SYSTEM

- Bleed the HCU . Refer to Section 206-00.
- Test drive the vehicle and verify vehicle tracking during an ABS stop.

Does the vehicle track correctly during an ABS stop?

Yes	The brake system is operating correctly. The concern may have been caused by air in the hydraulic system or a sticky valve.
No	GO to S3 .

S3 CHECK THE ABS INLET VALVE (CLOSED POSITION)

- Connect the scan tool.
- Ignition ON.
- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Rotate all the wheels to make sure they rotate freely (the transmission must be in NEUTRAL).
- Apply moderate brake pedal effort.
- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- Toggle the left front inlet valve (LF_INLET) active command ON and at the same time, have an assistant attempt to rotate the LF wheel.

Does the LF wheel rotate?

Yes	TOGGLE the LF_INLET active command OFF. INSTALL a new HCU . REFER to Hydraulic Control Unit (HCU) in this section. TEST the system for normal operation.
No	TOGGLE the LF_INLET active command OFF. GO to S4 .

S4 CHECK THE ABS INLET VALVE (OPEN POSITION)

- Apply moderate brake pedal effort.
- Have an assistant attempt to rotate the LF wheel.

Does the LF wheel rotate?

Yes	INSTALL a new HCU . REFER to Hydraulic Control Unit (HCU) in this section. TEST the system for normal operation.
No	GO to S5 .

S5 CHECK THE ABS OUTPUT VALVE (OPEN POSITION)

- Apply moderate brake pedal effort.
- Toggle the left front outlet valve (LF_OUTLET) and the LF_INLET active commands ON and, at the same time, have an assistant attempt to rotate the LF wheel.

Does the LF wheel rotate?

Yes	TOGGLE all active commands OFF. INSTALL a new HCU . REFER to Hydraulic Control Unit (HCU) in this section. TEST the system for normal operation.
No	TOGGLE all active commands OFF. GO to S6 .

S6 CHECK THE ABS INLET AND OUTLET VALVE (CLOSED POSITION)

- Apply moderate brake pedal effort.
- Have an assistant attempt to rotate the LF wheel.

Does the LF wheel rotate?

Yes	INSTALL a new HCU . REFER to Hydraulic Control Unit (HCU) in this section. TEST the system for normal operation.
No	REPEAT this procedure (GO to S3) for the RF, LR and RR wheels using the appropriate active commands. If no failure occurs, the system is operating normally.

Pinpoint Test T: DTCs U0151 and U0452

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module uses information received from several sensors (wheel speed sensors, steering wheel rotation sensor, stability/traction control system sensors, etc.) to determine if vehicle stability, traction or roll warrants ~~ESC~~ or ~~RSC~~® interaction. If an event is going to occur, the ABS module adjusts the brake torque at specific wheels and the PCM reduces engine torque to make the vehicle more stable. Once the ABS module has determined that the stability event has been corrected, it returns the brake and engine torque to normal and continues to monitor the vehicle. DTCs set in the ignition cycle can only be cleared by cycling the ignition off and on once the fault has been corrected. The ~~RCM~~ measures the yaw rate and vehicle acceleration, then sends the signal through a secondary, dedicated ~~CAN~~ to the ABS module. The ABS module uses this information for AdvanceTrac® and ~~RSC~~® system operation. The ABS module also compares this information to information it receives from other sensors (brake pedal input, steering wheel) to determine how valid the information is and to determine if an instability event is about to occur.

The RCM and the ABS module communicate along circuits VCA23 (BU/WH) and VCA24 (GN/OG). An open circuit or a short to power or ground on any one of these circuits will cause the ABS module to set a DTC. Also, if the RCM is not mounted in the correct position, with the electrical connector facing the driver-side of the vehicle, the ABS module will also set DTCs.

DTC Description	Fault Trigger Condition
- U0151 — Lost Communication with Restraints Control Module (RCM) - U0452 — Invalid Data Received from RCM	The RCM uses a dedicated communication bus to send information to the ABS module. If, during normal operation, the ABS module does not receive any information from the RCM for more than 250 milliseconds, DTC U0151 or U0452 will be set. U0151 monitors both the HS-CAN and the dedicated CAN circuits. If the ABS module does not receive the RCM serial number, over the HS-CAN , DTC U0151 sets. It also sets if the sensor data is not received over the dedicated CAN from the RCM .

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RCM
- ABS module

PINPOINT TEST T : DTCS U0151 AND U0452

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may cause damage to the connector.
NOTE: <i>Supplemental Restraint System (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .</i>
NOTE: <i>The SRS must be fully operational and free of faults before releasing the vehicle to the customer.</i>

T1 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
 - NOTE:** *The Network Test does not test the dedicated CAN communications between the ABS module and RCM .*
- Using a scan tool, carry out a Network Test.

Does the RCM pass the Network Test?

Yes	GO to T2 .
No	DIAGNOSE the no communication with the RCM . REFER to Section 418-00.

T2 RECHECK THE ABS MODULE DTCS

- Using a diagnostic scan tool, carry out the ABS module self-test.
- Clear the ABS module DTCs.
- Ignition OFF.
- Ignition ON.
- Using a diagnostic scan tool, carry out the ABS module self-test.
- Record the ABS module DTCs.

Is DTC U0151 or U0452 retrieved again?

Yes	GO to T3 .
No	If no DTCs are retrieved again, the system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition.

T3 REVIEW THE RECORDED ABS MODULE DTCS

- Review the previously recorded ABS module DTCs.

Was DTC B1676 recorded?

Yes	GO to Pinpoint Test A
No	GO to T4 .

T4 CHECK FOR RCM DTCS

- Using a diagnostic scan tool, carry out the RCM self-test.

Are any DTCs present in the RCM ?

Yes	DIAGNOSE the RCM DTCs. REFER to Section 501-20B. TEST the system for normal operation.
No	GO to T5 .

T5 CHECK FOR DTC U0151:XX OR U0452:XX SET IN OTHER MODULES

- Using a diagnostic scan tool, carry out the self-test for all modules.
- Retrieve and record all CMDTCs .

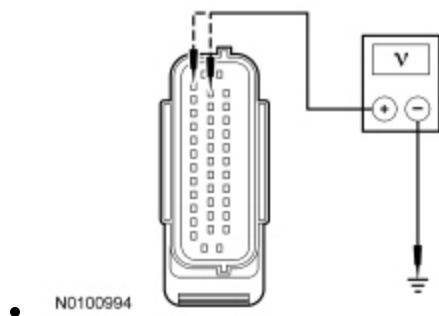
Is DTC U0151:xx or U0452:xx set in 1 or more modules in addition to the ABS module?

Yes	CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW the TSB instructions. If no TSBs address this concern, DIAGNOSE the RCM concern. Section 501-20B.
No	GO to T6 .

T6 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO VOLTAGE

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.
- Wait one minute then disconnect RCM [C310A](#) and [C310B](#).
- Disconnect ABS module [C135](#).
- Ignition ON.
- Measure the voltage between ground and:
 - ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.
 - ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.

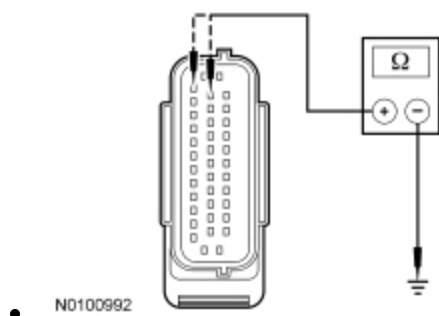


Is any voltage present?

Yes	REPAIR the circuit. CLEAR all CMDTCs . PERFORM a road test. REPEAT the self test.
No	GO to T7 .

T7 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between ground and:
 - ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.
 - ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.

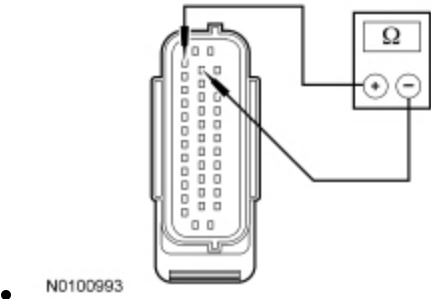


Is the resistance greater than 10,000 ohms?

Yes	GO to T8 .
No	REPAIR the circuit. CLEAR all <u>CMDTCs</u> . PERFORM a road test. REPEAT the self test.

T8 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side and ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.



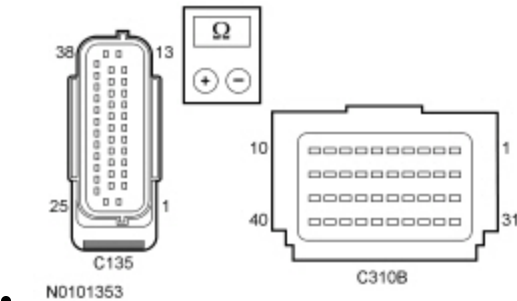
Is the resistance greater than 10,000 ohms?

Yes	GO to T9 .
No	REPAIR the circuit. CLEAR all <u>CMDTCs</u> . PERFORM a road test. REPEAT the self test.

T9 CHECK THE DEDICATED CAN CIRCUITS FOR AN OPEN

- Measure the resistance between the ABS module [C135](#), harness side and the ~~RCM~~ [C310B](#), harness side as indicated in the following chart:

ABS Module	Circuit	RCM
C135 Pin 37	VCA23 (BU/WH)	C310B Pin 20
C135 Pin 24	VCA24 (GN/OG)	C310B Pin 19



Are the resistances less than 5 ohms?

Yes	GO to T10 .
No	REPAIR the circuit. CLEAR all CMDTCs . PERFORM a road test. REPEAT the self-test.

T10 CHECK THE RCM FOR CORRECT OPERATION

- Ignition OFF.
- Inspect the ~~RCM~~ electrical connector, ABS module electrical connector and related in-line connectors.
- Repair:
 - corrosion (install new connector or terminal - clean module pins)
 - damaged or bent pins - install new terminals or pins
 - pushed-out pins - install new pins as necessary
- Connect ~~RCM~~ [C310A](#) and [C310B](#). Make sure they seat and latch correctly.
- Connect ABS module [C135](#) and related in-line connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

Is the concern still present?

Yes	CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW the TSB instructions. If no TSBs address this concern, DIAGNOSE the RCM concern. Section 501-20B.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues.

Pinpoint Test U: DTC C1278

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

When the ignition is ON and the battery voltage is greater than 10 volts, the ~~RCM~~ measures the yaw rate and sends the signal along a dedicated ~~CAN~~ to the ABS module. The ~~PSCM~~ broadcasts the steering angle over the ~~HS-CAN~~ . On vehicles equipped with a 6.2L engine, the ~~SCCM~~ broadcasts the steering angle, this information is used for vehicle stability control.

The ~~RCM~~ and the ABS module communicate along circuits VCA23 (BU/WH) and VCA24 (GN/OG). An open circuit or a short to power or ground on any one of these circuits causes the ABS module to set a DTC.

Once the cause of the concern has been repaired, the ignition must be cycled from ON to OFF and back to ON to clear the DTC from the ABS module.

DTC Description	Fault Trigger Condition
C1278 — Steering Wheel Angle 1 and 2 Signal Faulted	The ABS module uses the information received from wheel speed sensors and the yaw rate from the RCM to calculate the current steering angle and compares this information to the steering

DTC Description	Fault Trigger Condition
	angle messages received from the PSCM or SCCM , if the calculated steering angle is different than what is received, DTC C1278 is set.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Wheels or tires
- PSCM concern
- SCCM concern
- RCM concern
- Trailer concern
- ABS module concern

PINPOINT TEST U : DTC C1278

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may cause damage to the connector.
NOTICE: The ignition must be OFF, and remain OFF, when disconnecting PSCM electrical connectors. Failure to follow this direction may lead to DTCs being set in the PSCM which cannot be cleared, and result in the need to install a new EPAS steering gear.
NOTE: <i>Supplemental Restraint System (SRS) components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .</i>
NOTE: <i>The SRS must be fully operational and free of faults before releasing the vehicle to the customer.</i>
U1 CHECK FOR RETURNING ABS MODULE DTCS

- Ignition ON.
- Using a diagnostic scan tool, clear the ABS module DTCs.
- Ignition OFF.
- Ignition ON.
- Test drive the vehicle above 12 km/h (7 mph) and make at least 1 left turn and 1 right turn.
- Using a diagnostic scan tool, carry out the ABS module self-test.
- Record all ABS module DTCs.

Are any DTCs present in the ABS module?

Yes	If DTC C1278 is present, GO to U2 . For all other ABS module DTCs, GO to the ABS Module DTC Chart.
No	The fault is not present and cannot be recreated at this time. CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW the TSB instructions. If no TSBs address this concern, INSPECT all in-line connectors and the wire harness between the RCM and the ABS module for corrosion, loose terminals, spread terminals, loose wire connections, frayed wire connections, pinched wire harnesses and cut or pierced wire harnesses. REPAIR as necessary. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U2 CHECK THE WHEELS AND TIRES

- Inspect the wheels and tires for any damage or excessive wear.
- Verify all wheels and tires are the same size and match the size indicated on the [V.C](#) label.
- Verify the inflation pressure of all 4 tires is correct as indicated on the [V.C](#) label.

Are the wheels and tires OK?

Yes	GO to U3 .
No	INSTALL the correct size wheel or tire as necessary. ADJUST tire pressure as necessary. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U3 CHECK THE WHEEL ALIGNMENT

- Using alignment equipment and the manufacturer's instructions, check the wheel alignment.

Is the wheel alignment within specifications?

Yes	GO to U4 .
No	ALIGN the front wheels. REFER to Section 204-00. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U4 CHECK THE STEERING GEAR INSTALLATION

- Place the transmission in NEUTRAL and position the vehicle on a hoist. For additional information, Section 100-02.
- Inspect the steering gear bellows boots for any obvious signs of damage.
- Inspect the steering gear fasteners for looseness, cross-threading or signs of obvious damage.

Are the steering gear, bellows boots and fasteners OK?

Yes	For vehicles equipped with a 6.2L engine, GO to U6 . For all other vehicles, GO to U5
No	REPAIR or INSTALL new as necessary. REFER to Section 211-00A or Section 211-00B. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U5 VERIFY THE CORRECT STEERING GEAR IS INSTALLED

- Using a diagnostic scan tool, retrieve the **PSCM** part number and verify the vehicle has the correct **PSCM** installed.

Is the correct **PSCM installed in the vehicle?**

Yes	GO to U6 .
No	INSTALL a new EPAS. Refer to the appropriate Removal and Installation procedure in Section 211-02. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U6 CHECK FOR TRAILER TOW OPERATION

NOTE: A trailer incorrectly connected to the vehicle can cause the vehicle to constantly drift, requiring constant steering wheel input from the driver. This can cause DTC C1278 to set due to steering wheel angle, long-term offset compensation failure.

- Ask the customer if the concern happened while towing a trailer.

Did the concern happen while towing a trailer?

Yes	ADVISE the customer to review the Owner Information regarding trailer towing. If the trailer was connected and used correctly, GO to U7 .
No	GO to U7 .

U7 CHECK THE COMMUNICATION NETWORK

- Ignition ON.
- NOTE:** The Network Test does not test the dedicated **CAN** between the ABS module and **RCM**.

Using the diagnostic scan tool, carry out the Network Test.

Do the **RCM , **PSCM** (if equipped) and **SCCM** (if equipped) pass the Network Test?**

Yes	GO to U8 .
No	DIAGNOSE the RCM , PSCM or SCCM does not communicate with the diagnostic scan tool. REFER to Section 418-00. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U8 CHECK FOR RCM DTCS

- Using a diagnostic scan tool, carry out the **RCM** self-test.

Are any DTCS present in the **RCM** ?

Yes	DIAGNOSE all RCM DTCS. REFER to Section 501-20B. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCS.
No	For vehicles equipped with a 6.2L engine, GO to U10 . For all other vehicles, GO to U9 .

U9 CHECK FOR PSCM DTCS

- Using a diagnostic scan tool, carry out the **PSCM** self-test.

Are any DTCS present in the **PSCM** ?

Yes	DIAGNOSE all PSCM DTCS. REFER to Section 211-00A. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCS.
No	GO to U12 .

U10 CHECK FOR SCCM DTCS

- Using a diagnostic scan tool, carry out the **SCCM** self-test.

Are any DTCS present in the **SCCM** ?

Yes	DIAGNOSE all SCCM DTCS. REFER to Section 211-05. Once the concern has been repaired, CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCS.
No	GO to U11 .

U11 CHECK THE SCCM STEERING ANGLE (SW_ANGLE) PID

- Using a diagnostic scan tool, monitor the **SCCM** SW_ANGLE **PID** .
- Slowly turn the steering wheel from lock-to-lock and monitor the SW_ANGLE **PID** . The **PID** should increase by 1 degree at a time with a total of 780 degrees.

Does the **PID** value increase smoothly by 1 degree at a time with a maximum of 780 degrees?

Yes	GO to U12 .
No	INSTALL a new steering wheel rotation sensor. REFER to Section 206-09.

U12 VERIFY STABILITY CONTROL SENSOR CALIBRATION

- Using a diagnostic scan tool, carry out the IVD Initialization routine.
- Ignition OFF.
- Ignition ON.
- Test drive the vehicle above 12 km/h (7 mph) and make at least 1 left turn and 1 right turn.

Is the Advance Trac® warning indicator illuminated?

Yes	Using a diagnostic scan tool, CARRY OUT the ABS module self-test. If DTC C1278 is present, GO to U13 . For all other ABS module DTCs, GO to the ABS Module DTC Chart.
No	The condition which caused the concern is no longer present. The concern was most likely due to an incomplete calibration or a failed calibration of the stability control sensors. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U13 CHECK THE ABS MODULE PID VALUES

NOTE: The vehicle must be on level ground and at a complete standstill. Any vehicle movement results in false values for this test.

- Ignition OFF.
- Ignition ON.
- Using a diagnostic scan tool, monitor the following ABS module PIDs :
 - Yaw rate (YAW_RATE_2)
 - Lateral acceleration (LAT_ACCL_2)
 - Longitudinal acceleration (LONG_ACCL)
 - Roll rate (ROLLRAT)

Is the YAW_RATE_2 PID value between -3.5 and 3.5, the LAT_ACCL_2 PID value between -0.125 and 0.125, the LONG_ACCL PID value between -0.15 and 0.15, and the ROLLRAT PID value between -3.5 and 3.5?

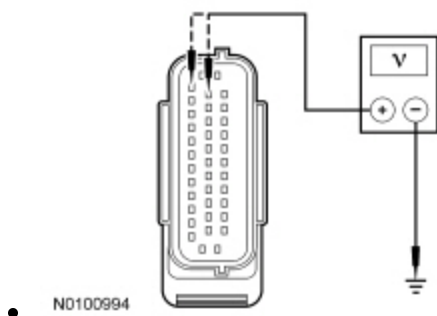
Yes	GO to U18 .
No	GO to U14 .

U14 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO VOLTAGE

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.
- Wait one minute then disconnect RCM [C310A](#) and [C310B](#).
- Disconnect ABS module [C135](#).
- Ignition ON.

- Measure the voltage between ground and:
 - ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.
 - ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.

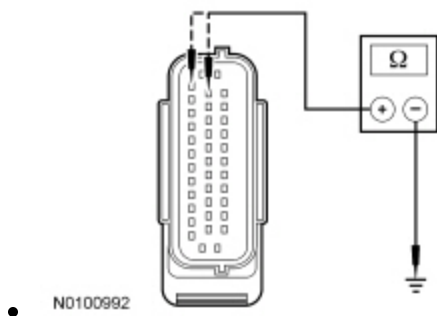


Is any voltage present?

Yes	REPAIR the circuit. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.
No	GO to U15 .

U15 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between ground and:
 - ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.
 - ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.

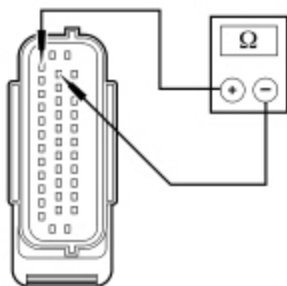


Is the resistance greater than 10,000 ohms?

Yes	GO to U16 .
No	REPAIR the circuit. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U16 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between ABS module [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side and ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side.



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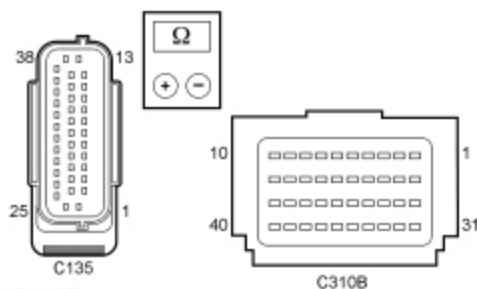
Is the resistance greater than 10,000 ohms?

Yes	GO to U17 .
No	REPAIR the circuit. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U17 CHECK THE DEDICATED CAN CIRCUITS FOR AN OPEN

- Measure the resistance between the ABS module [C135](#), harness side and the RCM [C310B](#), harness side as indicated in the following chart:

ABS Module	Circuit	RCM
C135 Pin 37	VCA23 (BU/WH)	C310B Pin 20
C135 Pin 24	VCA24 (GN/OG)	C310B Pin 19



N0101353

Are the resistances less than 5 ohms?

Yes	GO to U18 .
No	REPAIR the circuit. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U18 CHECK THE RCM INSTALLATION AND MOUNTING SURFACE

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.

- Wait one minute then disconnect **RCM** [C310A](#) and [C310B](#) (if not previously disconnected).
- Inspect the **RCM** installation and make sure the fasteners are fully seated and tightened to specifications. Refer to Restraints Control Module (RCM) Removal and Installation in Section 501-20B.
- Inspect the mounting surface for damage, corrosion or foreign material.

Is the **RCM installed correctly and is the mounting surface clean and free from damage?**

Yes	GO to U19 .
No	CLEAN or REPAIR the mounting surface or correctly REINSTALL the RCM as necessary, REFER to Section 501-20B. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U19 VERIFY ALL WIRING CONNECTIONS

WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Ignition OFF.
- Disconnect ABS module [C135](#) (if not previously disconnected).
- Wait one minute then disconnect **RCM** [C310A](#) and [C310B](#) (if not previously disconnected).
- Disconnect all wire harness, in-line electrical connectors between the ABS module and the **RCM** .
- Using a good light source, inspect **all** disconnected electrical connectors for the following:
 - corrosion - install new connector or terminal and clean the module pins.
 - damaged or bent pins - install new terminals or pins.
 - pushed-out pins - install new pins as necessary.
 - spread terminals - install new terminals as necessary.

Are the connectors free of corrosion, damaged pins, bent pins, pushed-out pins and spread terminals?

Yes	GO to U20 .
No	REPAIR the connector or terminals. Refer to Wiring Diagrams Cell 5 for schematic and connector information. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.

U20 CHECK THE FOR CORRECT RCM OPERATION

- Connect ABS module [C135](#). Make sure it seats and latches correctly.
- Connect **RCM** [C310A](#) and [C310B](#). Make sure they seat and latch correctly.
- Connect all wire harness, in-line electrical connectors between the ABS module and the **RCM** . Make sure they seat and latch correctly.
- Test drive the vehicle above 12 km/h (7 mph) and make at least 1 left turn and 1 right turn.

Is the Advance Trac® warning indicator illuminated?

Yes	CHECK QASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW the TSB instructions. If no TSBs address this concern, DIAGNOSE the RCM concern. Section 501-20B. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.
No	The system is operating correctly at this time. The concern may have been caused by a wiring harness or electrical connector issue. ADDRESS the root cause of any wire harness or connector issues. CYCLE the ignition from ON to OFF and back to ON. CLEAR all ABS module DTCs.